AmarVote

Secure Cryptographic Voting Platform

Complete System Guide

This guide covers the full end-to-end workflow of AmarVote:

- Platform Setup & Authentication
- Election Creation & Configuration
- Guardian Key Ceremony
- Encrypted Voting & Benaloh Challenge
- Tallying & Cryptographic Decryption
- Independent Ballot Verification

ElectionGuard Powered   End-to-End Verifiable

Version 1.0 | June 18, 2026

Contents

About This Guide

This document is the authoritative, comprehensive reference for the **AmarVote** platform. It is written for a general audience — no prior knowledge of cryptography, election administration, or software engineering is required.

AmarVote is built on the **ElectionGuard** specification developed by Microsoft, which guarantees *end-to-end verifiability*: every voter can independently confirm that their ballot was cast as intended, recorded as cast, and counted as recorded — without ever exposing how they voted.

Note

This guide deliberately mirrors the actual workflow of the application in the exact order you would encounter it. Read sequentially for first-time use, or jump to a specific chapter if you need a quick reference.

Roles at a Glance

Role	Responsibilities
Owner	Full platform control: manages the authorised-user list, assigns roles, and sets global permissions.
App Admin	A platform-level administrator who can create and manage elections on AmarVote. There can be an unlimited number of App Admins on the platform.
Election Admin	The specific user who created a particular election; responsible for configuring, scheduling, and overseeing that election. Each election has exactly one Election Admin. Note: "Election Admin" is simply a term used to identify the creator of a specific election — the user's platform role remains App Admin (or Owner).
Guardian	Participates in the cryptographic key ceremony; holds a share of the decryption key needed to tally results.
Voter	Registers, receives election access, casts an encrypted ballot, and independently verifies the result.

Note

Permission hierarchy: Owner > App Admin > User (voter). The Owner can elevate or restrict any account. All roles require a registered account. An **Election Admin** is the creator of a specific election — there is exactly one Election Admin per election — but this does not change their platform-level role (App Admin or Owner).

AmarVote Workflow Overview

Before you begin the detailed chapters, the following diagrams provide a high-level map of the complete AmarVote workflow. Use them as a quick reference to see how each phase connects — from platform setup and election creation through voting, tallying, and verification.

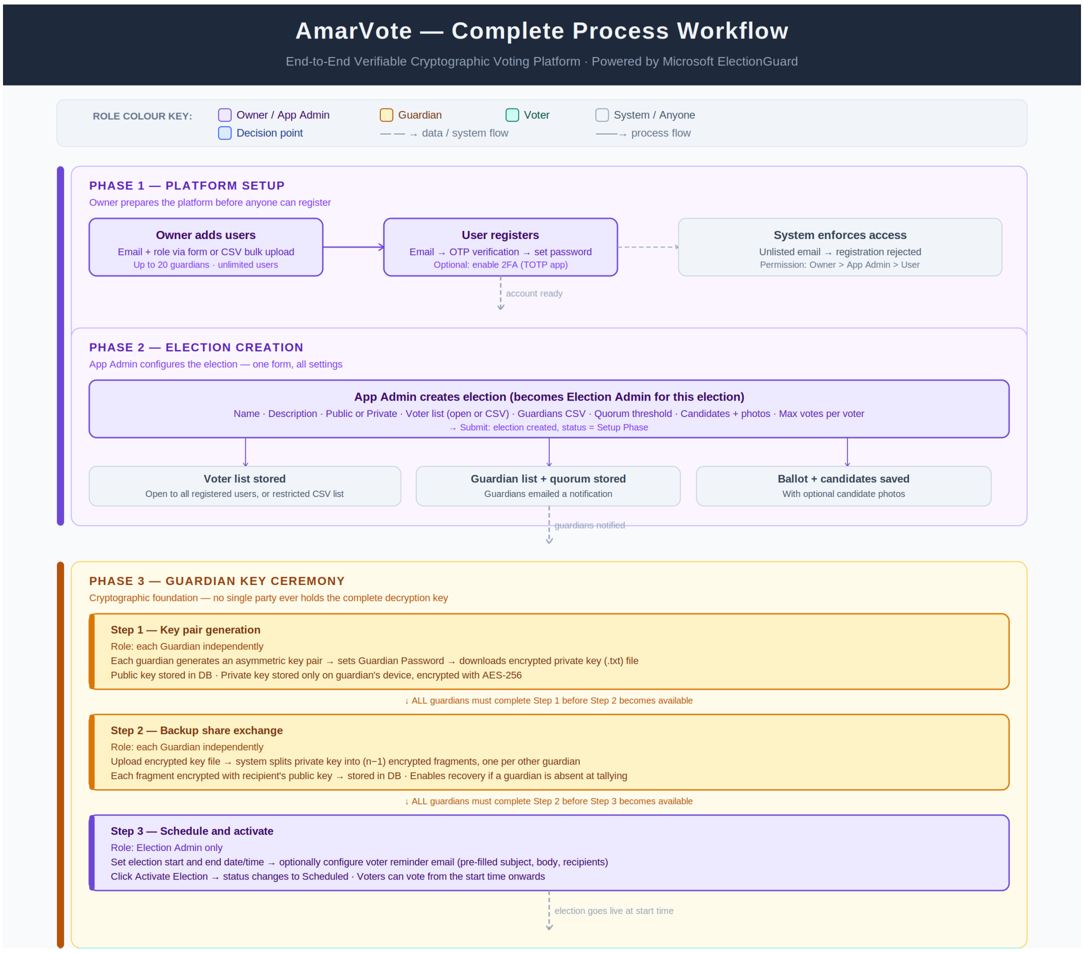


Figure 1: AmarVote end-to-end workflow — Part 1: platform setup, election creation, and guardian key ceremony

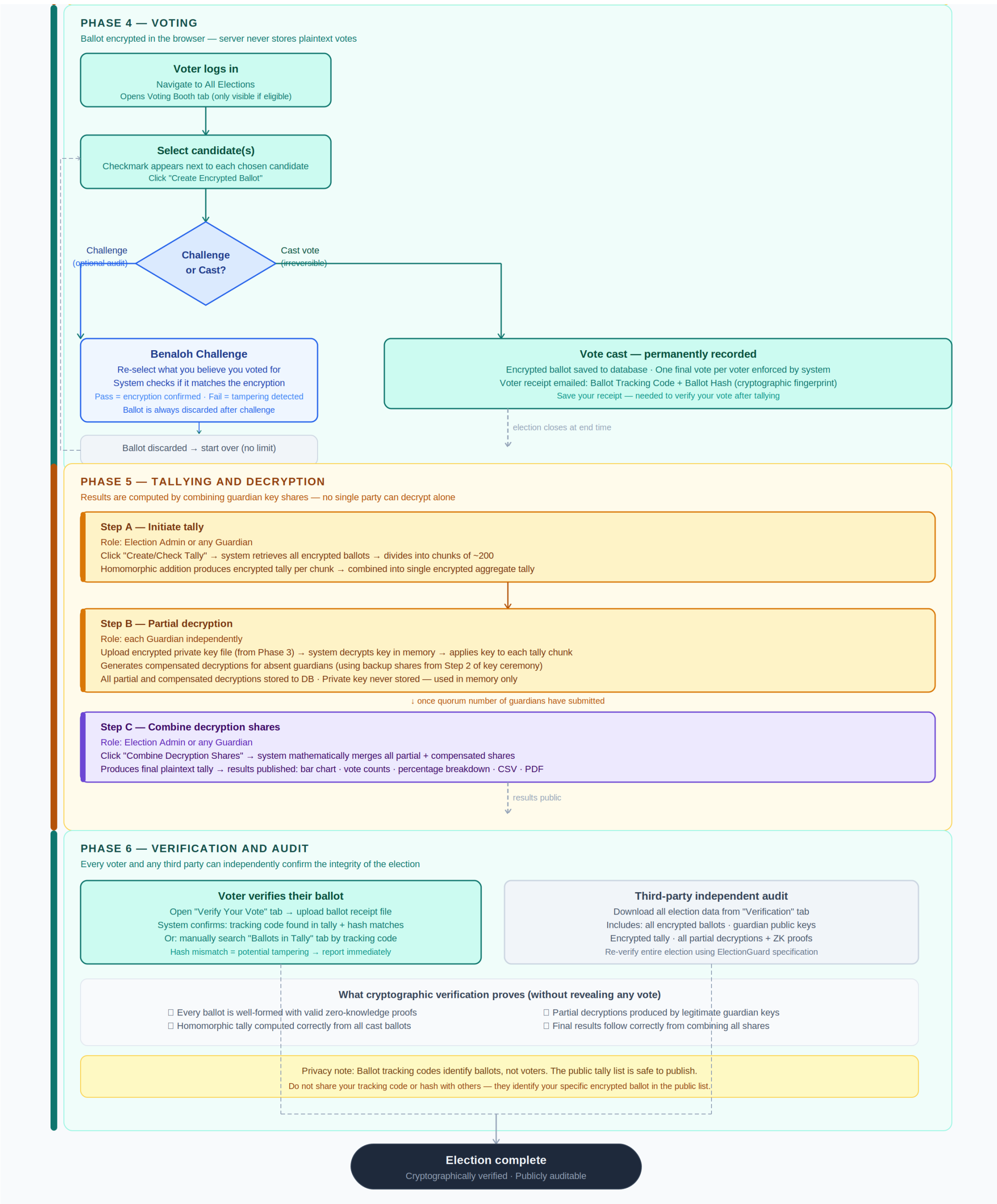


Figure 2: AmarVote end-to-end workflow — Part 2: voting, tallying, decryption, and verification

Part I

Platform Foundation

Chapter 1

Access Control & Authorisation System

AmarVote is a closed platform — not just anyone can register. Before a user can even create an account, their email address must appear on the **Authorised User List** managed by the Owner. This design ensures that elections are only accessible by vetted participants.

1.1 The Authorised User List

The Owner maintains a registry of permitted email addresses. Each entry in the list specifies:

- The user's `email address`
- The `assigned role` (Owner, App Admin, or User)

1.1.1 Adding Individual Users

1. Log in to AmarVote with an Owner account.
2. Open the navigation menu by clicking the menu icon at the top of the sidebar.

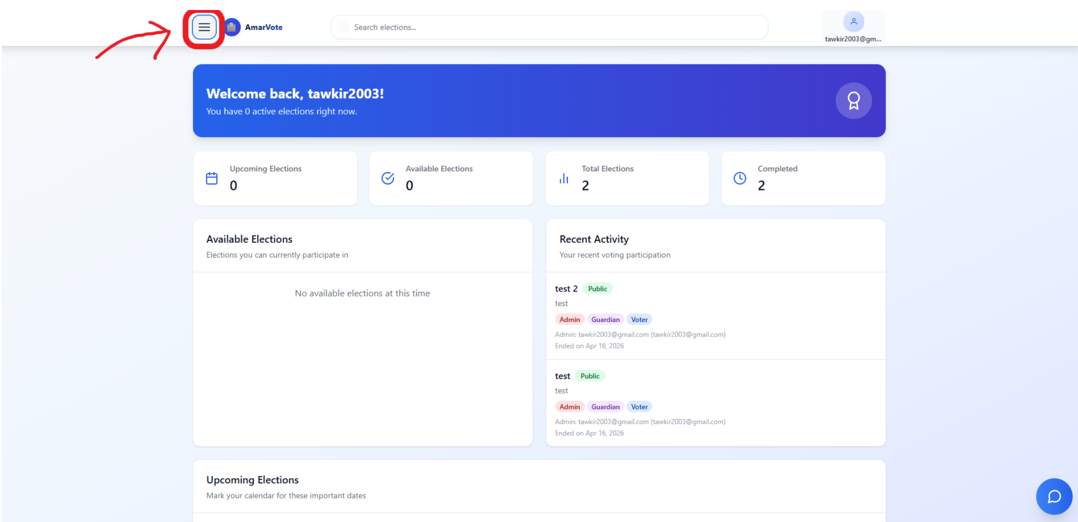


Figure 1.1: Opening the navigation menu

3. Navigate to **Authenticated Users** in the sidebar.

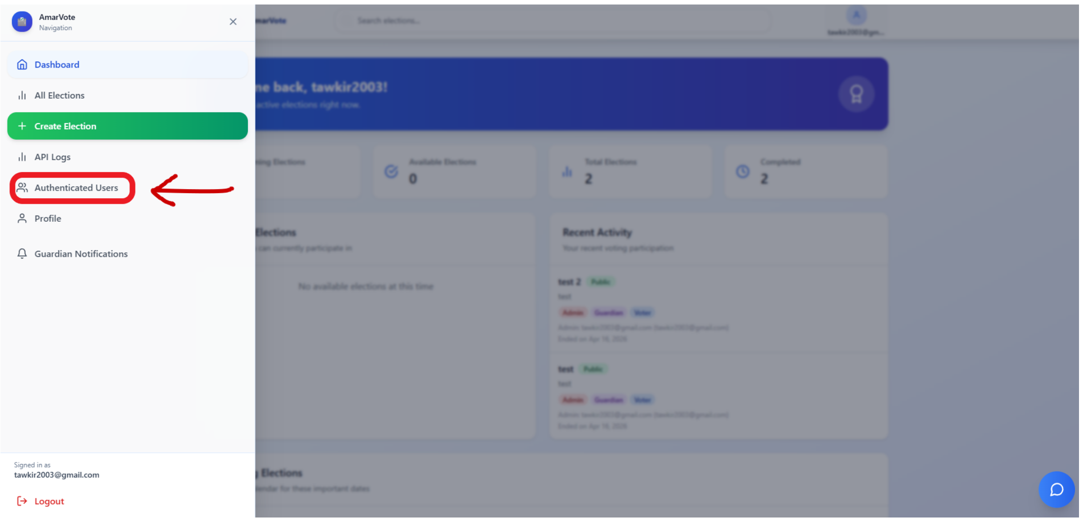


Figure 1.2: Selecting *Authenticated Users* from the navigation menu

4. Enter the email address in the `Add User Email` field.
5. Select the appropriate role from the `Role` dropdown.
6. Click **Add**.

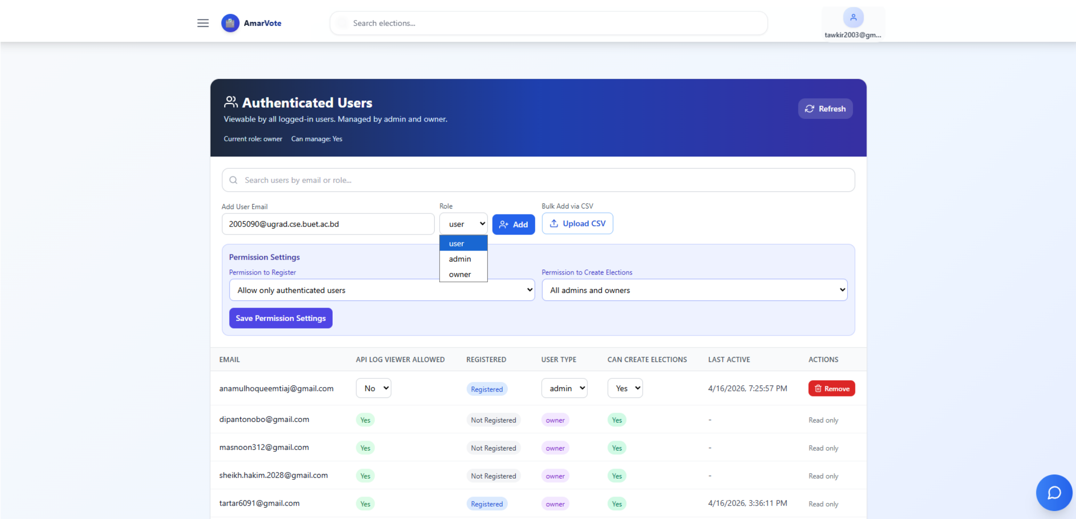


Figure 1.3: Adding an individual user to the Authorised User List

> **Tip**
>
> If you are adding a small number of users (fewer than 10), the individual entry method is faster. For larger groups, use CSV bulk upload.

1.1.2 Bulk Upload via CSV

For adding many users at once:

1. Prepare a `.csv` file with one column, emails

Figure 1.4: Example CSV format for bulk user upload

2. Navigate to **Authenticated Users** → **Upload CSV**.
3. Click and select your prepared CSV file.

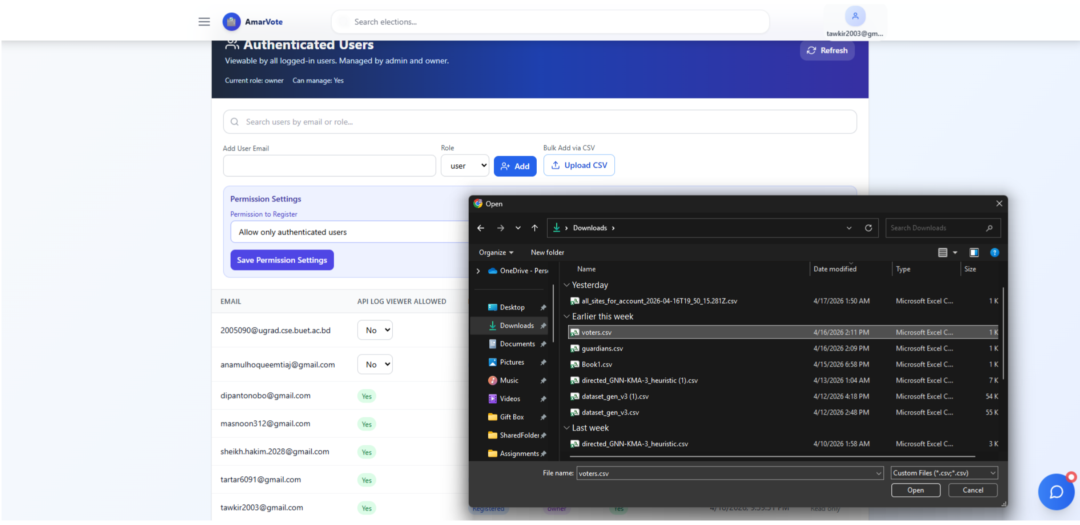


Figure 1.5: Bulk importing users via CSV upload

> **Important**
>
> Duplicate email entries in the CSV will be ignored. If an email already exists in the system, it will be ignored, not duplicated.

1.2 Role Capabilities Summary

Capability	Owner	Admin (App)	User
Make anyone owner	✓	✗	✗
Demote any owner	✓	✗	✗
Manage authorised-user list	✓	✓	✗
Assign roles	✓	✓	✗
Create elections (default)	✓	✓	✗
Create elections (if granted)	✓	✓	✓
Vote in elections	✓	✓	✓
View public elections	✓	✓	✓
View private elections (if invited)	✓	✓	✓

Note

Election creation setting: The Owner can enable a global setting allowing *all* users (not just App Admins) to create elections. When this setting is active, both App Admins and regular Users will see the **Create Election** button in their menu.

Chapter 2

Account Registration

Before using any feature of AmarVote, every participant — voter, guardian, or App Admin — must create a personal account. The registration process is a guided, multi-step flow that takes only a few minutes and involves email verification followed by an optional two-factor authentication setup.

2.1 Prerequisites

> **Important**
>
> You can only register if your email address has been added to the Authorised User List by an Owner. If you attempt to register with an unlisted email, the system will reject your request before any account is created.

2.2 Registration Workflow Overview

The full registration journey follows this sequence:

Home → Register → Enter Email → Code Sent → Enter Code → Set Password → Account Created

Each step is described in detail below, with screenshots corresponding to exactly what you will see on screen.

2.3 Step-by-Step Registration

2.3.1 Step 1 — The AmarVote Home Page

1. Open your web browser and navigate to the AmarVote application URL provided by your organisation.
2. You will arrive at the **AmarVote home page**.

Figure 2.1: The AmarVote home page — your entry point to the platform

3. Click the **Register** button, or navigate directly to `/register` in your browser's address bar.

2.3.2 Step 2 — Submit Your Email Address

1. The registration form will open, prompting you for your email address.

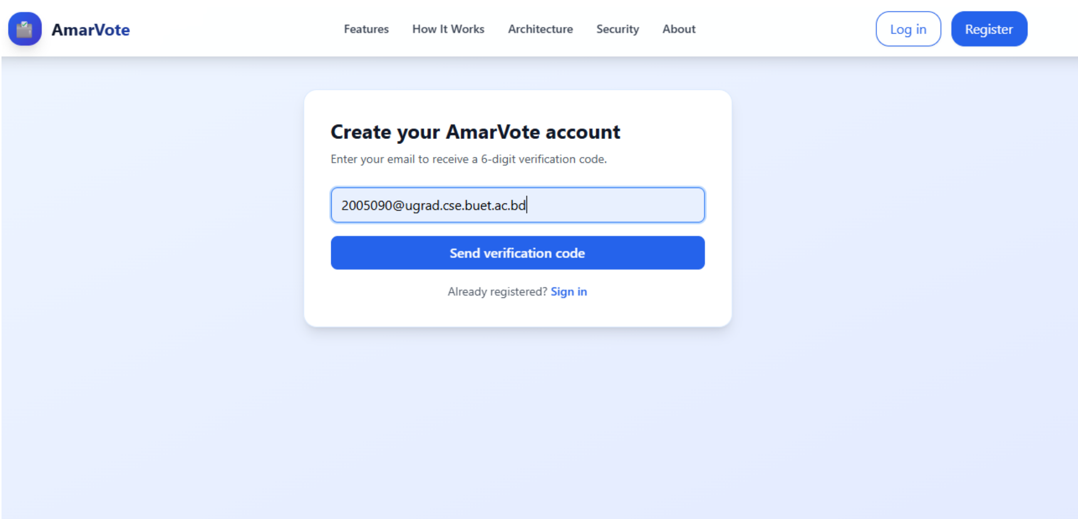


Figure 2.2: The registration page — enter the email address that was added to the Authorised User List

2. Enter your `Email Address` in the provided field. This must **exactly match** the email address the Owner or Admins added to the Authorised User List.

3. Click **Send Verification Code**.
4. The system will dispatch a **One-Time Password (OTP)** to your inbox and display a confirmation screen.

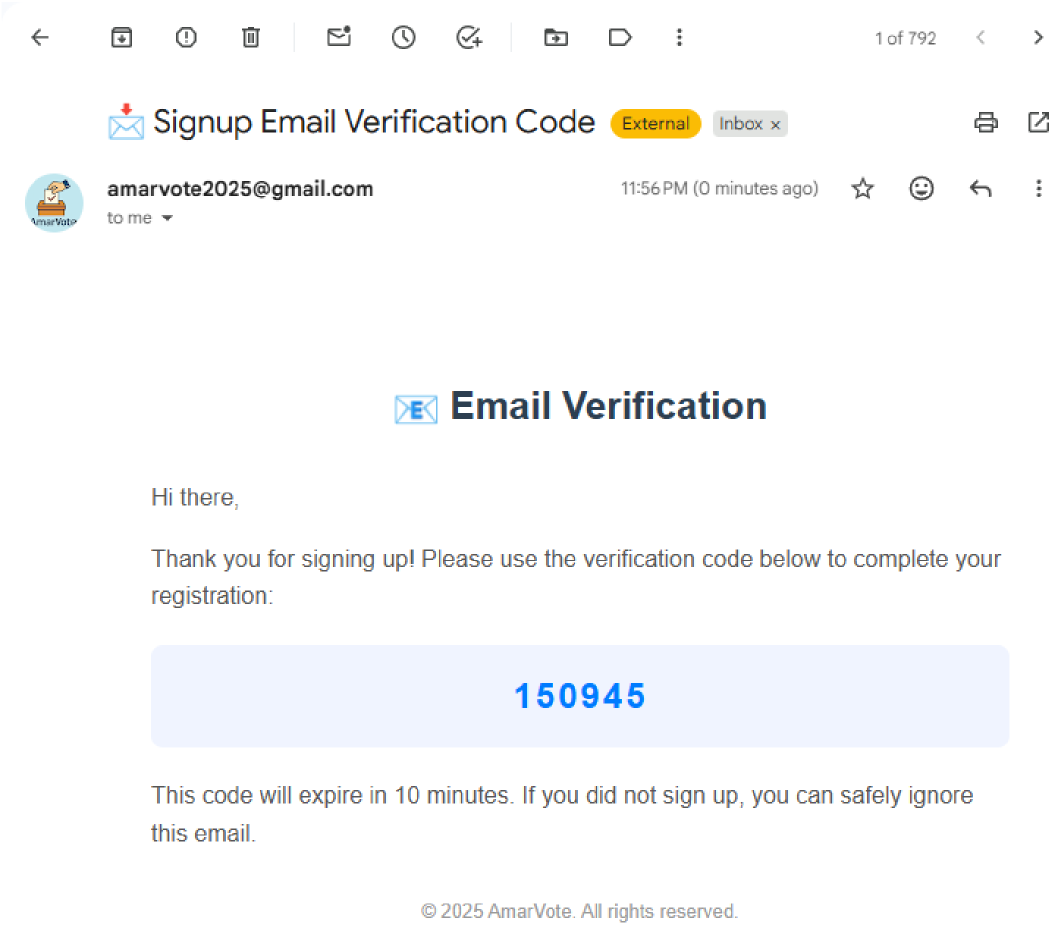


Figure 2.3: The verification code has been sent to your email

Tip

Check your spam or junk folder if you do not receive the email within two minutes. The OTP is typically valid for **10 minutes**. If it expires, return to the registration page and request a new code.

2.3.3 Step 3 — Enter the Verification Code

1. Open the email from AmarVote and locate the **6-digit verification code**.
2. Return to the registration page (it may still be open in your browser).
3. Enter the code in the `Verification Code` field.

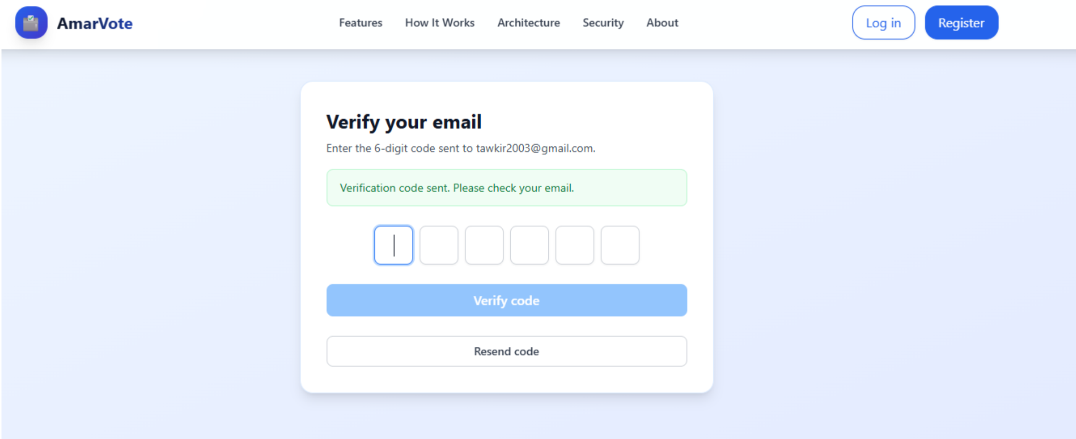


Figure 2.4: Entering the 6-digit OTP received by email to verify your address

4. Click **Verify Code**.
5. If the code is correct and has not expired, you will be automatically advanced to the password setup step.

Note

The verification code is single-use. Once entered successfully, it cannot be reused. If you enter an incorrect code, request a new one from the registration page.

2.3.4 Step 4 — Set Your Password

1. The password creation screen will appear.

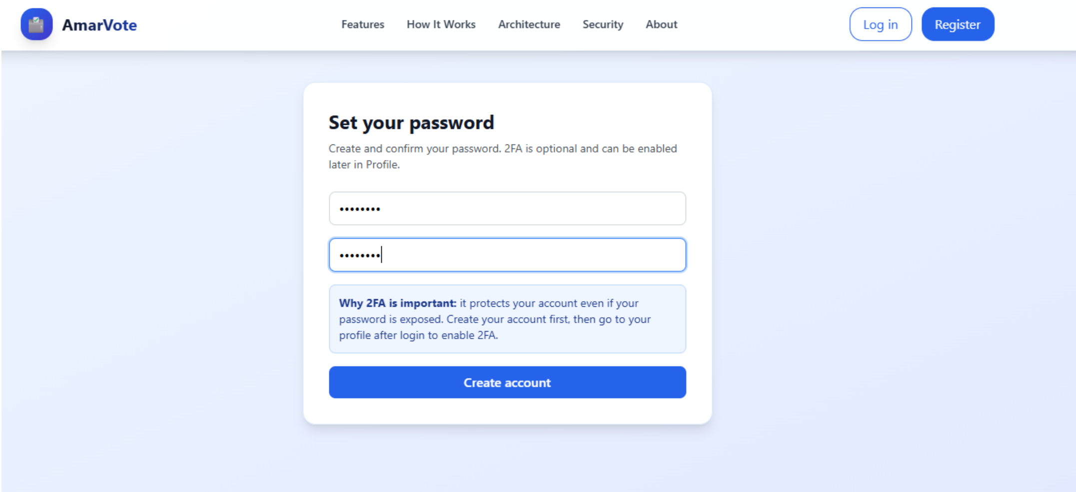


Figure 2.5: The password setup screen — choose a strong password for your AmarVote account

2. Enter a strong password in the `Password` field. A strong password contains at least **8 characters** and mixes uppercase letters, lowercase letters, numbers, and symbols.

3. Re-enter the identical password in the `Confirm Password` field.

4. Click **Create Account**.

5. Your account will be created and you will be redirected to the login page (or automatically signed in, depending on your platform configuration).

> **Warning**
>
> **Never share your password.** AmarVote staff will never ask for your password. Store it securely in a trusted password manager such as Bitwarden, 1Password, or your browser's built-in vault.

2.4 Two-Factor Authentication (2FA) — Optional but Strongly Recommended

After your account is created, you can enable Two-Factor Authentication (2FA) using **Google Authenticator** or any TOTP-compatible app (such as Authy or Microsoft Authenticator). 2FA adds a critical second layer of security — even if someone learns your password, they cannot log in without your physical device.

2.4.1 Enabling 2FA

1. Log in to your new account.

2. Click your profile in the top-right corner.

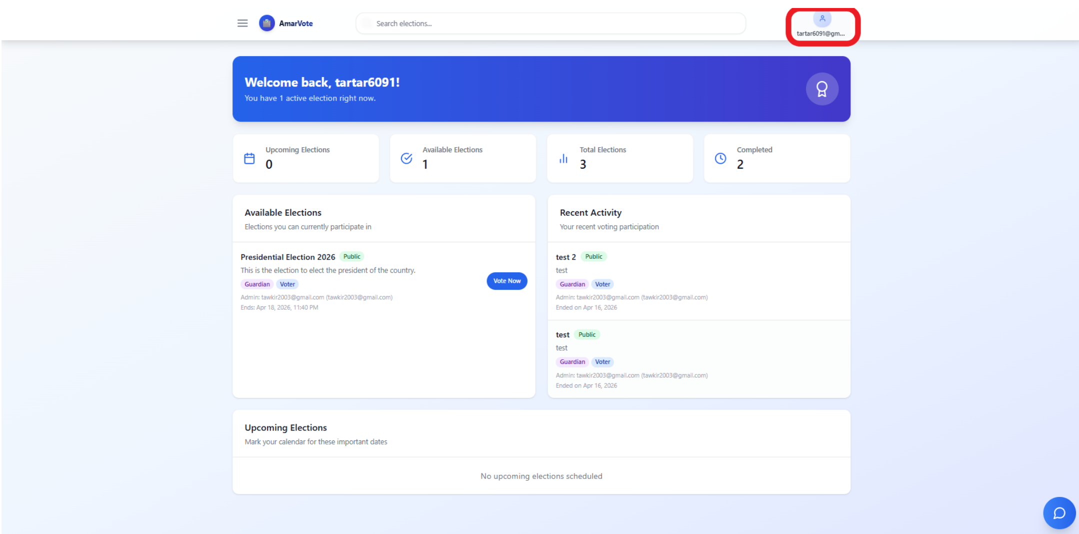


Figure 2.6: Accessing the profile menu from the top-right corner

3. Click **Turn on** Two-Step Verification. The 2FA setup screen will appear.

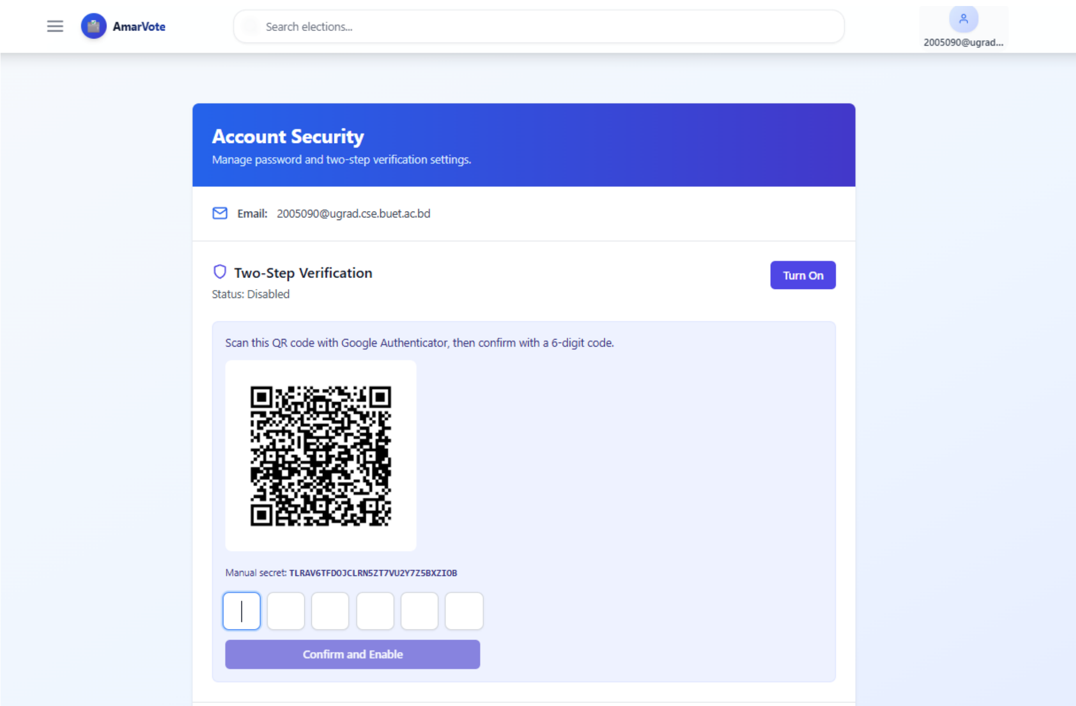


Figure 2.7: The Two-Factor Authentication setup screen, displaying the QR code to scan with your authenticator app

4. Open the Google/Microsoft Authenticator app on your smartphone.
5. Tap the + button and select **Scan a QR code**.
6. Point your camera at the QR code displayed on screen.
7. Your authenticator app will register AmarVote and begin displaying a rotating **6-digit code**.
8. Enter the current code in the `Verification Code` field on AmarVote and confirm.
9. Successful input will activate 2FA on your account immediately.

> **Important**
>
> **Safeguard your authenticator access.** If you lose access to the device running your authenticator app, you may be permanently locked out of your AmarVote account. Consider storing a backup code in a secure location, or using an authenticator app that supports encrypted cloud backup.

Chapter 3

Logging In

Once your account has been created, logging in to AmarVote is a straightforward process. The login flow begins from the same home page used for registration.

3.1 Login Workflow Overview

Home → Login → Enter Credentials → Dashboard

If Two-Factor Authentication is enabled, an additional step is inserted after credential entry:

Home → Login → Enter Credentials → Enter 2FA Code → Dashboard

3.2 Standard Login

1. Open your web browser and navigate to the AmarVote application URL. You will arrive at the **AmarVote home page**.

Figure 3.1: The AmarVote home page — click *Login* to proceed to the sign-in screen

2. Click **Login** on the home page, or navigate directly to `/login`.

3. The **login screen** will appear.

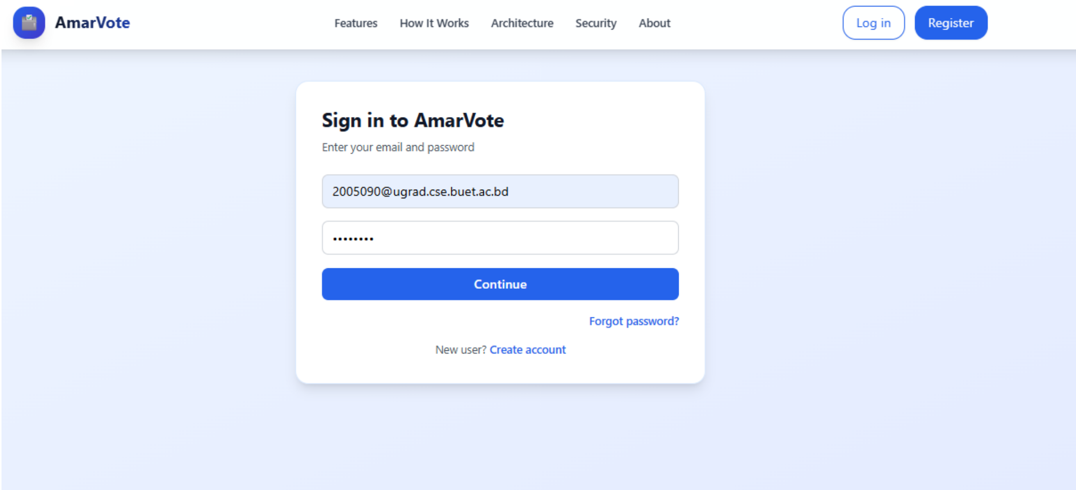


Figure 3.2: The AmarVote login screen — enter your registered email address and password

4. Enter your registered `Email Address`.

5. Enter your `Password`.

6. Click **Continue**.

7. Upon successful authentication, you will be redirected to your **Dashboard**.

3.3 Login with Two-Factor Authentication

If you have 2FA enabled on your account, the login flow includes one additional verification step after your credentials are accepted:

1. Complete Steps 1–5 of the Standard Login above (navigate to home, go to login screen, enter email and password).

2. Click **Continue**.

3. A prompt will appear asking for your `Authenticator Code`.

4. Open the Google Authenticator (or equivalent TOTP) app on your smartphone and copy the current **6-digit code** displayed for AmarVote.

5. Enter the code in the prompt and click **Verify**.

6. Upon successful verification, you will be redirected to your Dashboard.

Note

TOTP codes rotate every **30 seconds**. If the code expires while you are typing, simply wait for the next code to appear in your authenticator app and use that one instead.

3.4 Forgot Password

If you have forgotten your password, AmarVote provides a secure self-service reset flow:

1. On the login screen, click **Forgot Password?**.
2. Enter your registered `Email Address` and click **Send Verification Code**.
3. Check your inbox for an email containing a **password reset verification code** (valid for a limited time).
4. Enter the verification code on the reset page, then choose and confirm your new password.
5. Once set, you may log in immediately with your new credentials.

Tip

If the reset email does not arrive within a few minutes, check your spam or junk folder. The reset code is time-limited — if it expires, return to the login page and request a fresh one.

Part II

Election Administration

Chapter 4

Creating an Election

This chapter is intended for App Admins (and Users with election-creation permission). It walks through every option available when setting up a new election. The user who creates an election automatically becomes its **Election Admin** — the sole administrator responsible for managing that specific election.

> **Important**
>
> The **Create Election** button will only appear in your navigation menu if you have been granted election-creation permission. If you cannot see it, contact AmarVote Owner or Admin.

4.1 Accessing the Election Creation Form

1. Log in to AmarVote.
2. Open the navigation menu by clicking the menu icon at the top of the sidebar.

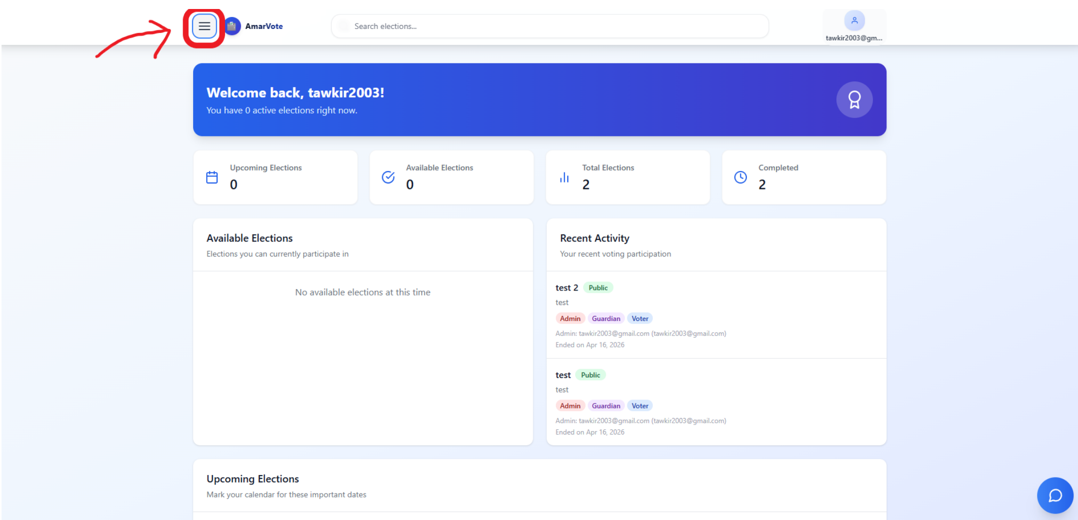


Figure 4.1: Opening the navigation menu to access election management options

3. In the left navigation sidebar, click **Create Election**.

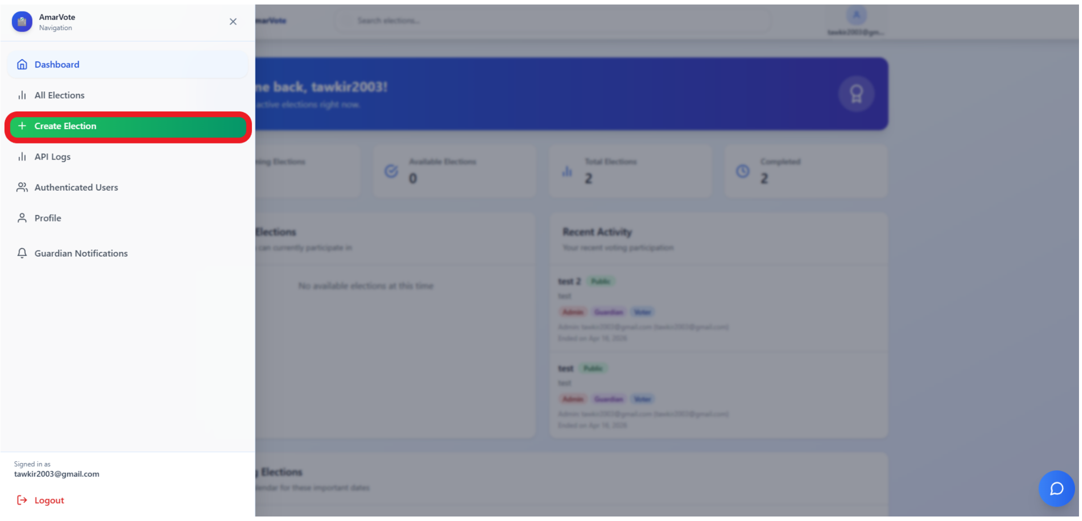


Figure 4.2: Selecting *Create Election* from the navigation menu

4. The **New Election** form will open.

4.2 Basic Election Information

4.2.1 Election Name & Description

- `Election Name`: A short, descriptive title (e.g., "Student Council Election 2025").
- `Description`: A longer explanation of the election's purpose, visible to voters.

4.2.2 Visibility: Public vs. Private

Setting	Effect
Public	All registered users can see the election and its results.
Private	Only invited voters, assigned guardians, and the Election Admin can view the election, its ballot, and its results.

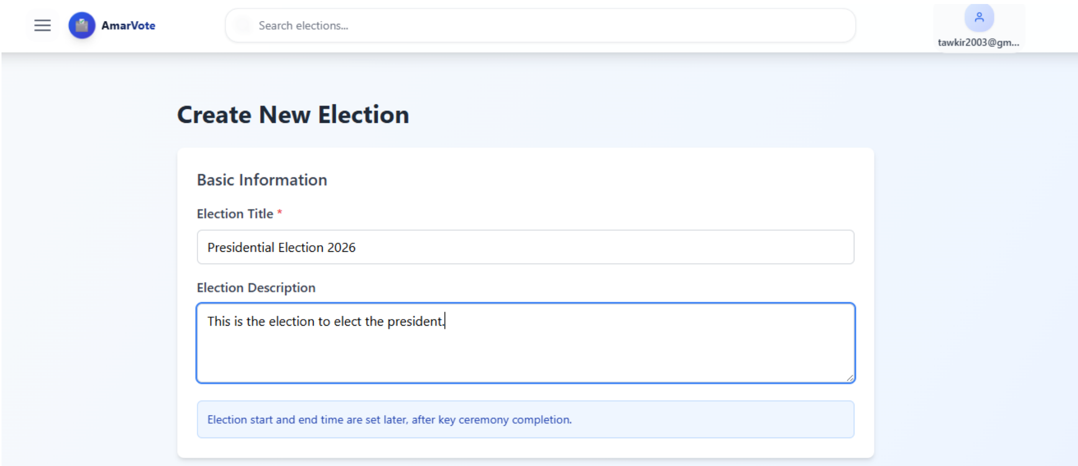


Figure 4.3: Election creation form — title, description, and maximum choice configuration

4.3 Voter Configuration

4.3.1 Open Voter List

If you want *any* registered AmarVote user to be able to vote:

1. Select **Anyone (All Registered Users)** from the `Voter List` dropdown.

4.3.2 Restricted Voter List (CSV Upload)

To limit voting to specific individuals:

1. Prepare a CSV file with one column: `just email addresses`.
   Example: `voter1@example.com`, one per row.

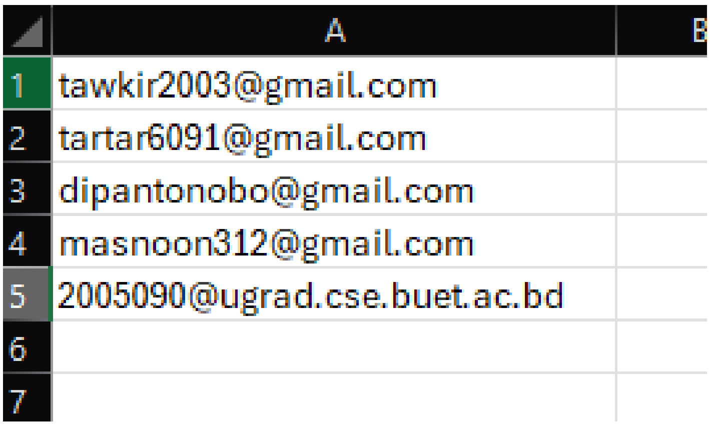


Figure 4.4: Example voter CSV file format — one email address per row

2. Click **Upload CSV** and select your file.
3. Review the voter count displayed after upload.

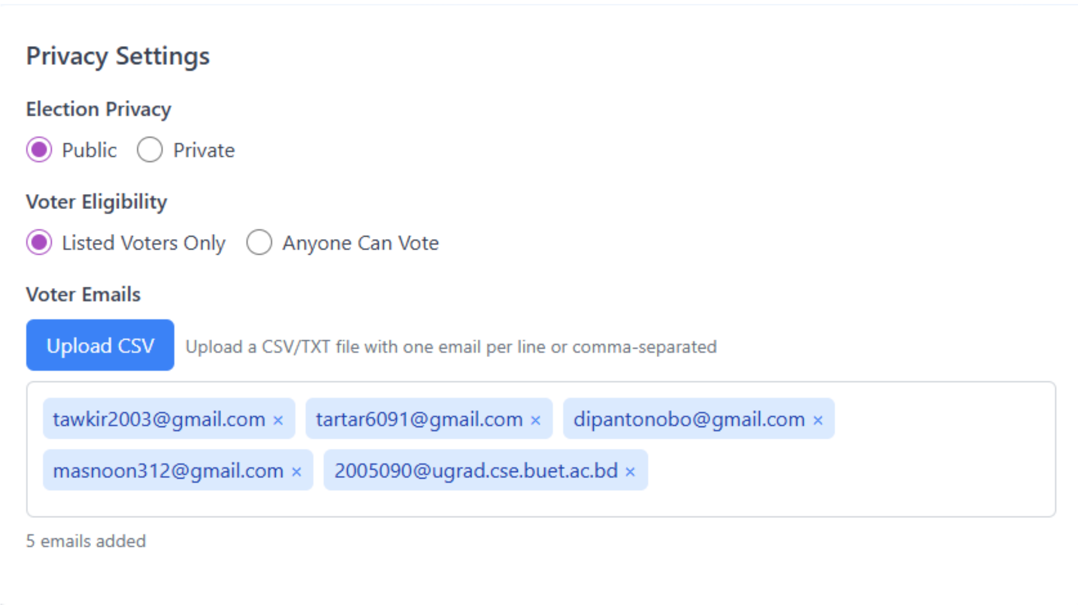


Figure 4.5: Uploading the restricted voter list during election creation

Note

Voters on the list who do not yet have an AmarVote account will need to register before they can vote. It is recommended to notify voters in advance.

4.4 Guardian Configuration

Guardians are trusted individuals who each hold a share of the cryptographic key used to decrypt the tally. Without a sufficient number (the *quorum*) of guardians cooperating, the results cannot be decrypted — ensuring no single party can tamper with or prematurely reveal results.

4.4.1 Uploading the Guardians List

1. Prepare a CSV file with one column: `just email addresses` (one guardian email per row).

Figure 4.6: Example guardians CSV file format — one guardian email per row

2. Click **Choose File** and select your file.
3. The system will display the total number of guardians detected.

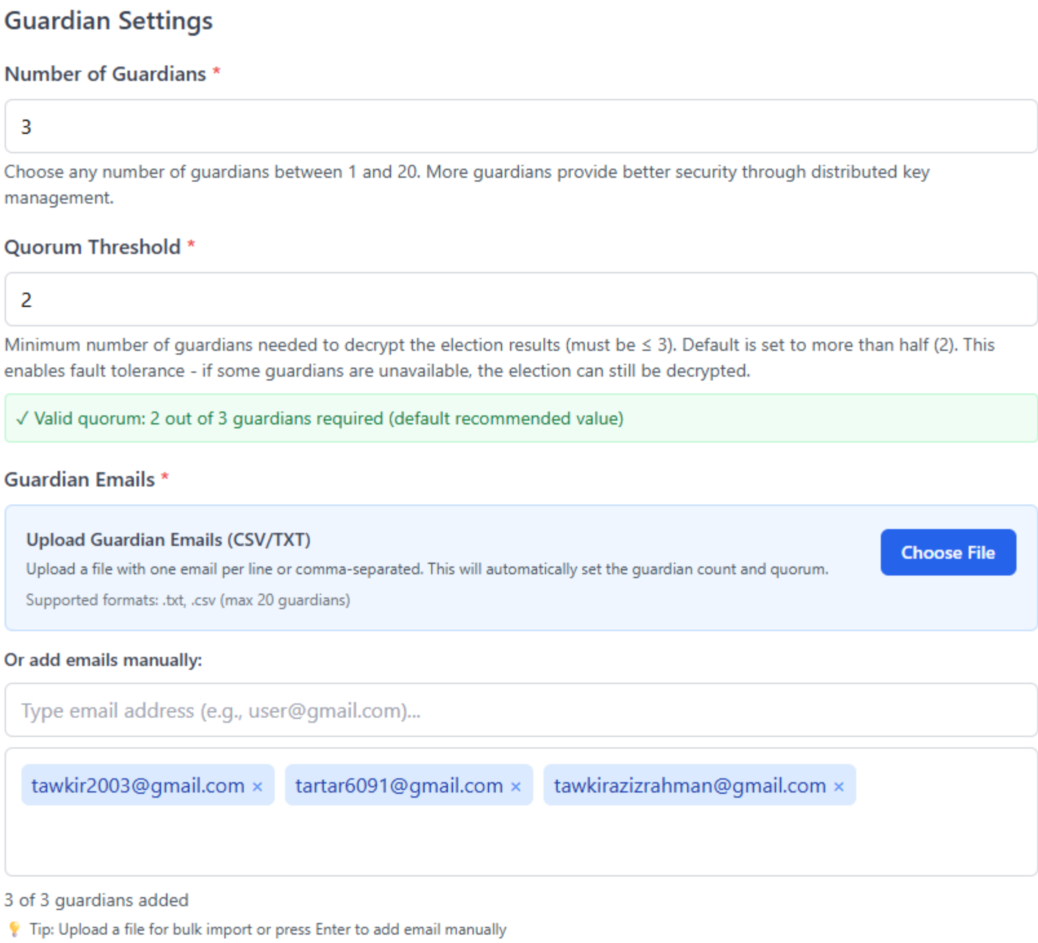


Figure 4.7: Adding guardians during election creation

4.4.2 Setting the Quorum

The **quorum** is the minimum number of guardians who must cooperate to decrypt the tally. This is a critical security parameter.

1. In the `Quorum` field, enter a number between 1 and the total number of guardians.
2. Recommended: set quorum to at least $\lfloor n/2 \rfloor + 1$ where n is the total guardian count.

> **Important**
>
> **Quorum Example:** If you have 5 guardians and set quorum to 3, any 3 guardians can decrypt the tally even if 2 guardians are unavailable. If quorum is set to 5, *all* guardians must be present — one absence blocks tallying permanently.

4.5 Ballot Configuration

4.5.1 Adding Candidates

For each candidate:

1. Click **Add Candidate**.
2. Enter the `Candidate Name`.
3. Upload a `Candidate Photo` (optional, JPG/PNG, max 10 MB).
4. Repeat for each candidate.

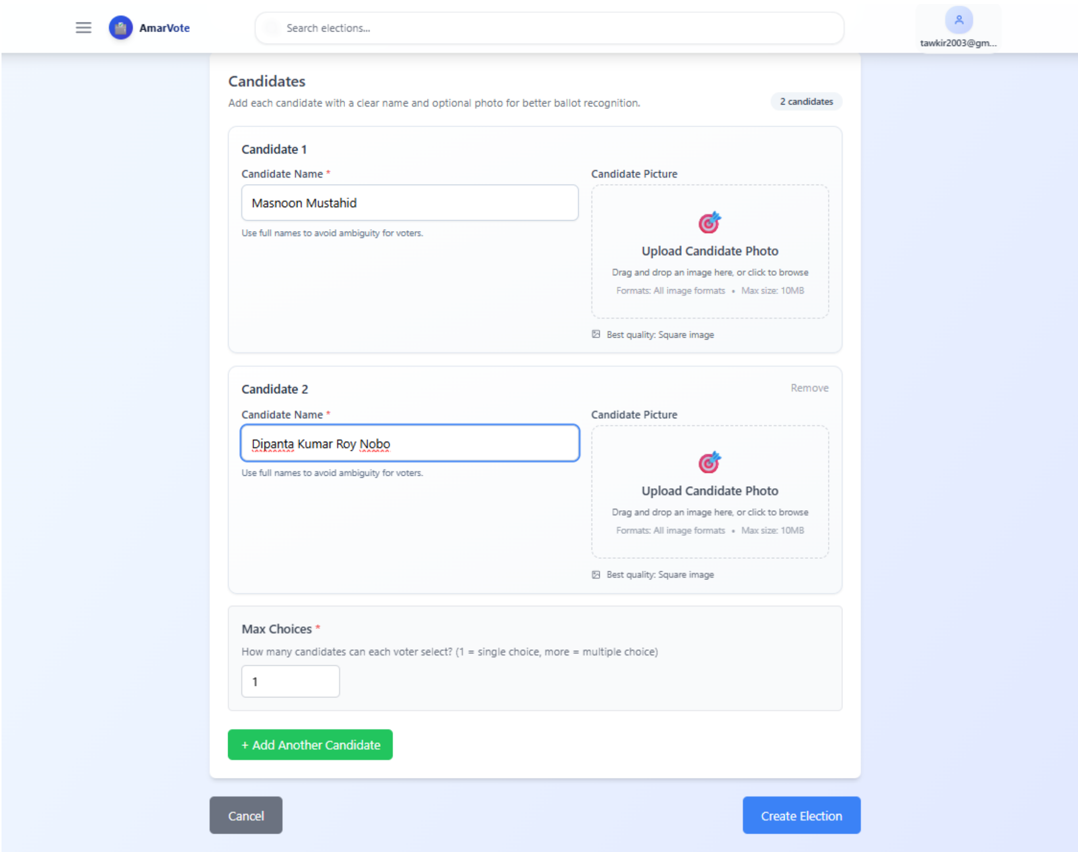


Figure 4.8: Adding candidates to the election ballot

Tip

Candidate photos significantly improve the voter experience, especially in large elections with many candidates. Providing them is strongly recommended.

4.5.2 Maximum Votes Per Voter

By default, each voter may select exactly one candidate. To allow multiple selections:

1. Locate the `Maximum Votes` field.
2. Enter the maximum number of candidates a voter may select (e.g., 3 means a voter can vote for 1, 2, or 3 candidates).

Note

Setting Maximum Votes to 1 creates a traditional single-choice election. Setting it to the total number of candidates allows approval voting.

4.6 Finalising Election Creation

After filling in all required fields:

1. Review all the information given in the page.
2. Click **Create Election**.
3. The system will provision the election and redirect you to the election's management page.
4. The election will now show as **Setup Phase** — the guardian key ceremony must be completed before the election can go live.

Chapter 5

The Guardian Key Ceremony

The **Key Ceremony** is the cryptographic heart of AmarVote. It is a multi-phase process in which each guardian generates, shares, and verifies cryptographic key material. The purpose is to ensure that the election's decryption key is *never* held by any single person — it is split and distributed in a mathematically provable way.

> **Note**
>
> **Why does this matter?** Even the election administrator cannot decrypt the ballots alone. A minimum of *quorum* guardians must cooperate, making unilateral manipulation of results mathematically impossible.

5.1 Phase 1 — Key Pair Generation

5.1.1 Guardian Actions — Phase 1

Each guardian must complete Phase 1 independently:

1. Log in to AmarVote.
2. You will see a notification indicating you have been assigned as a guardian.

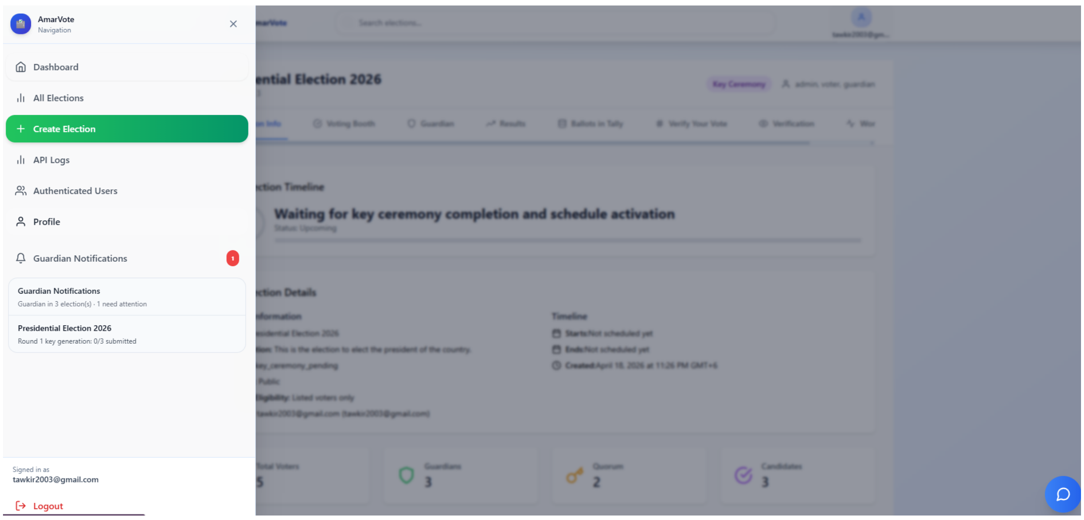


Figure 5.1: Guardian notification upon being assigned to an election

3. Navigate to **All Elections** and find the election. Open the election page and click

the **Guardian** tab.

4. You will see that **Phase 1: Key Pair Generation** is active.

5. Click **Generate Credentials** as many times as you like.

6. Every time you click, the system will cryptographically generate a new **asymmetric key pair** (a public key and a private key) specifically for this election.

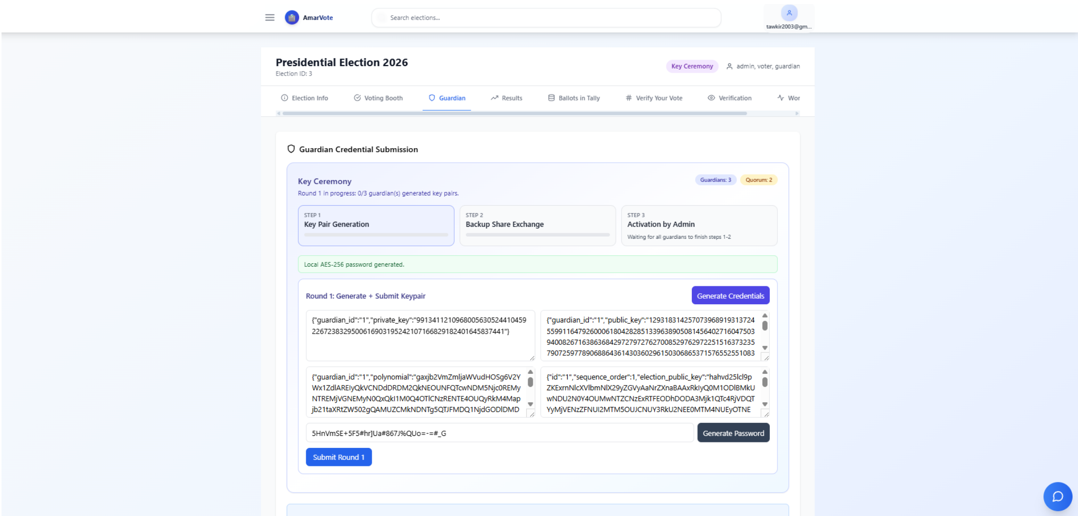


Figure 5.2: Phase 1 key pair generation interface

7. You will be prompted to create a **Guardian Password**. This password is used to encrypt your private key using AES-256 (a military-grade symmetric encryption standard).

8. **Important:** Click **Generate Password** to let the system generate a random, strong password for you, *or* enter your own password in the field.

9. Click **Submit Round 1**.

10. Your browser will automatically **download an encrypted private key file** (`.txt`) to your computer. **You must allow downloading that file, else the election can not go forward.**

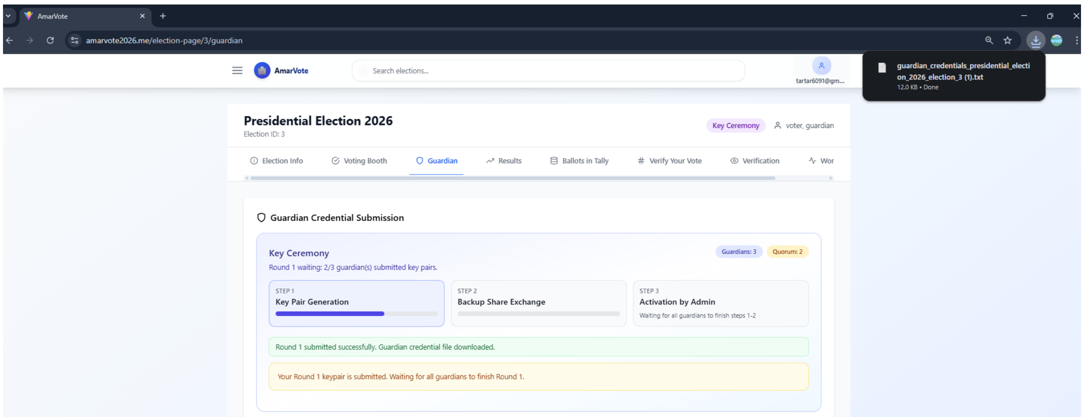


Figure 5.3: Downloading the encrypted private key file after key pair generation

11. Your **public key** is stored in the AmarVote database. Your **private key** is stored *only* on your device, encrypted with your guardian password.

> **Warning**
>
> **Critical — Do not lose the downloaded private key file**. If you lose that, you will not be able to participate in decryption. Depending on the quorum, this could prevent the election results from ever being decrypted.

> **Tip**
>
> Store the private key file in a secure location — an encrypted USB drive, a secure cloud vault, or your system's encrypted home directory. You don't need to store the guardian password, as it is stored in database using ML KEM 1024 post quantum encryption.

5.1.2 Phase Transition Condition

Phase 2 will not begin until **every** guardian assigned to the election has successfully completed Phase 1. The election page will show a progress indicator displaying how many guardians have submitted their public key.

5.2 Phase 2 — Backup Share Generation

Once all guardians have completed Phase 1, Phase 2 begins automatically. Each guardian must now generate **backup shares** — encrypted fragments of their private key that other guardians can use for disaster recovery.

5.2.1 How Backup Shares Work (Conceptual Overview)

Suppose there are 5 guardians (G1, G2, G3, G4, G5):

- Guardian G1 splits their private key into 4 fragments (one for each of the other 4 guardians).
- Each fragment is encrypted with the *public key* of the corresponding recipient guardian.
- For example, G1's fragment for G2 is encrypted so that *only G2* (using their private key) can decrypt it.
- These encrypted fragments are submitted to and stored in the database.

If G1 later loses their private key, any *quorum* number of other guardians can each provide their encrypted fragment. Together, the quorum can mathematically reconstruct G1's private key, allowing the tally to proceed.

5.2.2 Guardian Actions — Phase 2

1. Navigate to the election's **Guardian** tab.
2. You will see that **Phase 2: Backup Share Generation** is now active.
3. Click **Choose File** and select the `.txt` file you downloaded in Phase 1.
4. Click **Generate Backup Shares**.
5. The system will:
   (a) Decrypt your private key in memory.
   (b) Split it into $(n-1)$ encrypted fragments (one per other guardian).
   (c) Encrypt each fragment with the respective guardian's public key.
   (d) Store all encrypted fragments to the database.
6. Click **Submit Round 2** to upload the generated shares to the database.
7. A confirmation message will appear when Phase 2 is complete for you.

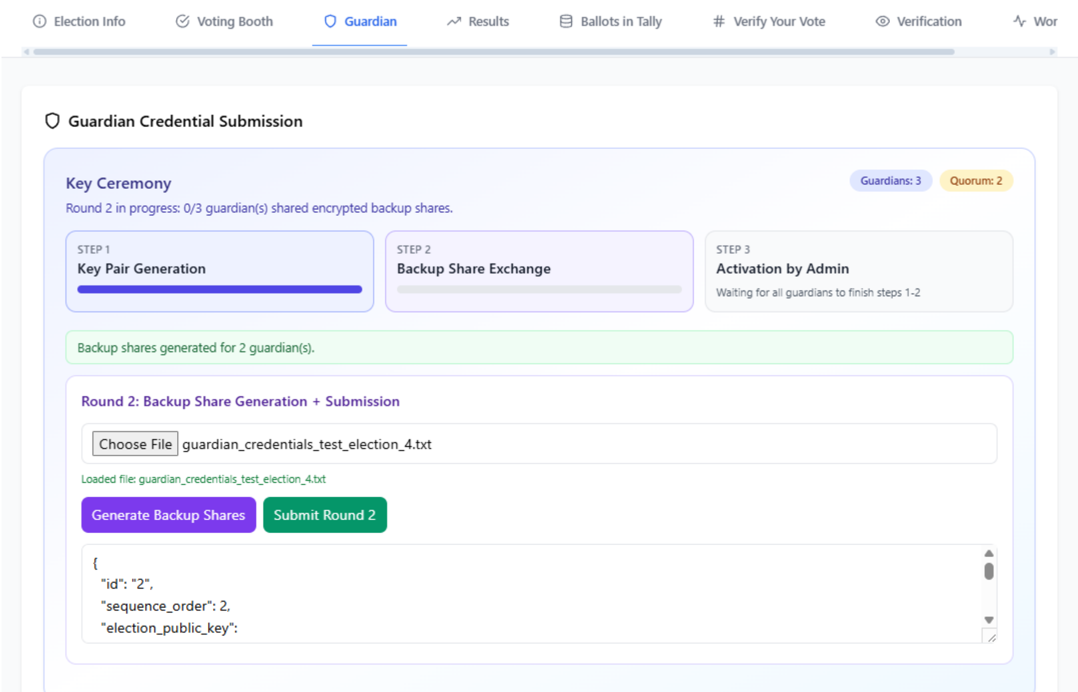


Figure 5.4: Phase 2 backup key share generation and submission

Important

Your decrypted private keys are used only in-memory during this step. They are **never stored** in the database. The database only stores the encrypted backup shares.

5.2.3 Phase 2 Transition Condition

Phase 3 begins only when **all** guardians have submitted their backup shares.

5.3 Phase 3 — Election Scheduling & Notifications (Election Admin)

When all guardians have completed Phase 2, the Election Admin is notified and Phase 3 becomes available. This phase allows the Election Admin to set the election schedule, and optionally configure and send a professional reminder email to voters.

Note

Election Admin reminder: The term *Election Admin* refers specifically to the user who created this election. There is exactly one Election Admin per election. All other platform-level App Admins do not have administrative access to elections they did not create.

5.3.1 Election Admin Actions — Phase 3

1. Log in with the Election Admin account (the account that created this election).
2. Navigate to the election's **Guardian** tab.
3. You will see the **Phase 3: Activation By Admin** panel is now active.

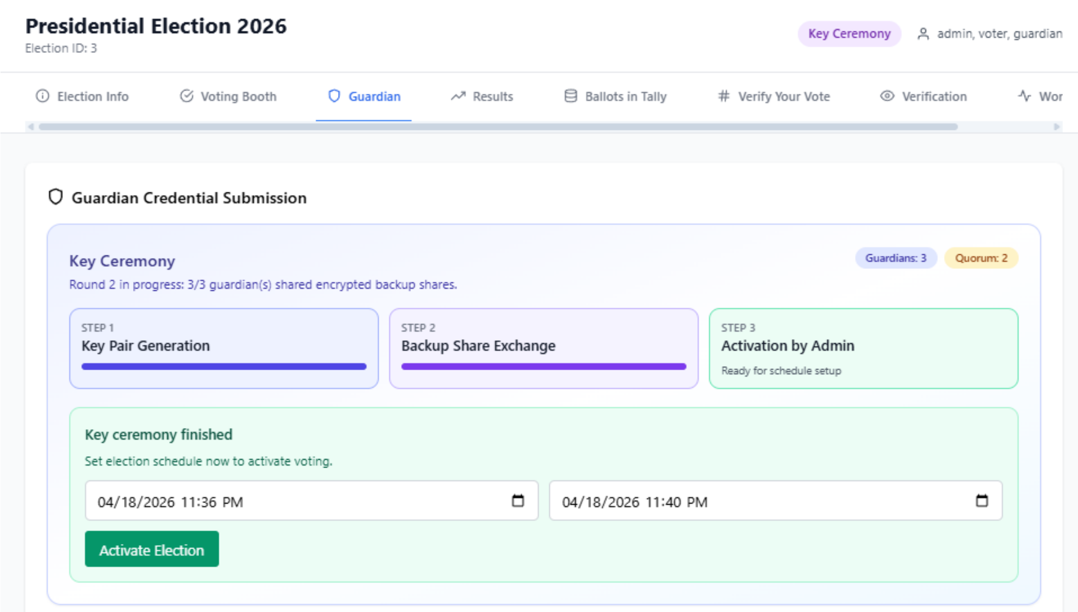


Figure 5.5: Phase 3 panel — election scheduling and optional voter notification configuration

Step A — Setting the Election Start and End Times

The scheduling interface features a **date-and-time picker** designed for ease and precision:

1. Click the `Election Start Date & Time` field. A calendar widget will appear, allowing you to:
   - Navigate months using the **left** and **right arrow** buttons.
   - Click any **date** to select it. The selected date highlights immediately.
   - Use the **hour** and **minute** scroll wheels (or type directly) to set the exact time. AM/PM toggle is available for 12-hour format, or the field accepts 24-hour format automatically.
2. Repeat the same process for the `Election End Date & Time` field.
3. The system will validate that the end time is after the start time. An error message will appear immediately if the selection is invalid.

> **Important**
>
> **Choose election times carefully.** Once the election is activated, the start and end times cannot be changed. Ensure you have accounted for your voters' time zones and any logistical constraints before proceeding.

Step B — Configuring a Voter Reminder Email (Optional)

Below the scheduling fields, you will find the **Send Reminder Email** section:

1. **Enable the Reminder Email:** Check the `Send Reminder Email to Voters` checkbox. When checked, the full reminder configuration panel expands beneath it.
2. **Email Receivers:**
   - By default, **all voters on the election's voter list** are pre-populated as recipients. A scrollable list displays each recipient's email address, as there may be hundreds or thousands of voters.
   - Each recipient entry has a × **(remove) button** on its right side. Click it to remove that specific voter from the reminder list.
   - At the bottom of the receiver list, an **Add Receiver** field with an **Add** button allows you to manually add any additional email address (e.g., a supervisor or observer not on the voter list).
3. **Email Subject:**
   The `Subject` field is **pre-filled with a professional subject line**, for example:
   "Reminder: [Election Title] — Voting is Now Open"
   You may edit this to suit your context.
4. **Email Body:**
   The `Message Body` text area is **pre-filled with a complete, professional email body** that automatically includes:
   - The **election title**.
   - The **election description**.
   - The **direct link** to the election page on AmarVote.
   - The **voting start and end times**.
   - A closing note encouraging voters to reach out if they have questions.

   You may freely edit the body text. The election link and other dynamic fields will be substituted with real values when the email is sent.
5. **Reminder Send Time:**
   Use the `Reminder Time` date-and-time picker to select *when* the reminder emails

will be sent. This uses the same modern, interactive picker as the start/end time fields:

- Select the date using the calendar widget.
- Set the hour and minute using the scroll wheels or direct input.
- The reminder time must be **before the election end time**.
- The reminder time can be set *before* the election start time (e.g., to notify voters in advance) or *during* the election period.

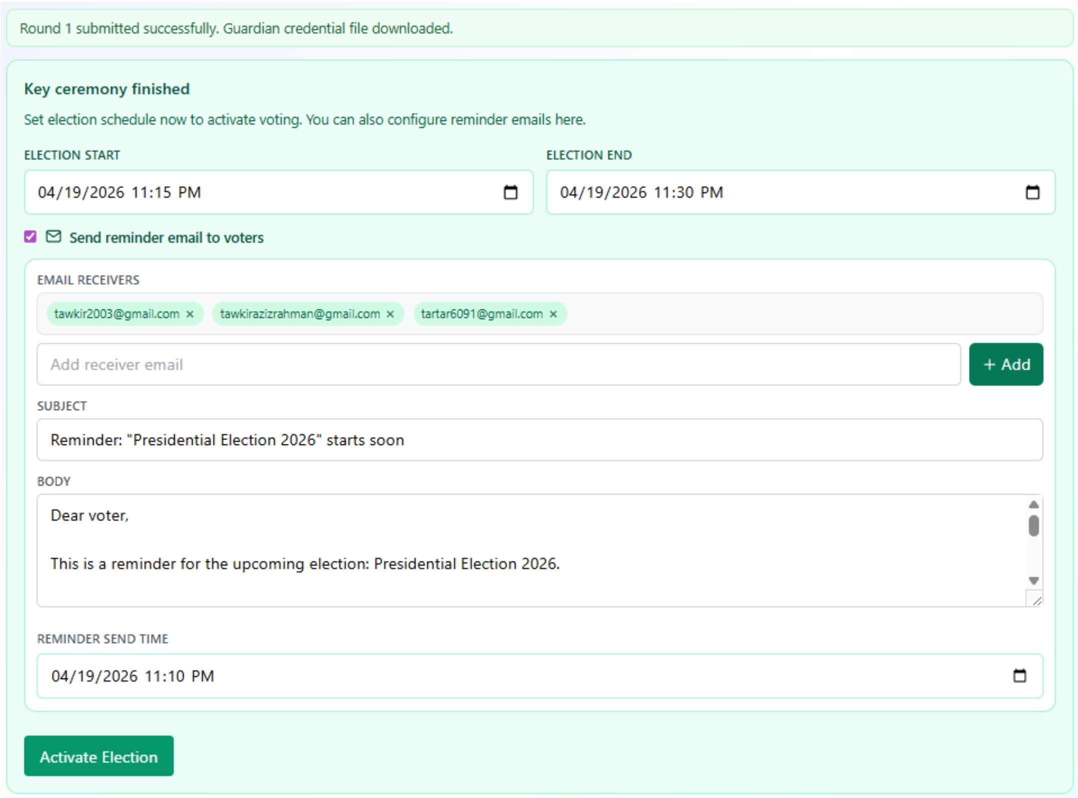


Figure 5.6: Voter reminder email configuration — receivers list with scroll and remove/add controls, pre-filled subject and body, and reminder send time picker

Tip

The default email body is professionally written and ready to send without modification. However, personalising it — for example, adding context about why the election matters or instructions specific to your organisation — will significantly improve voter engagement and turnout.

Note

If you do *not* wish to send a reminder, simply leave the `Send Reminder Email to Voters` checkbox unchecked. The scheduling step will still be completed normally when you activate the election.

Step C — Activating the Election

1. After setting the start and end times (and optionally configuring the reminder email), click **Activate Election**.
2. The election status changes to **Scheduled**. Voters will be able to vote starting from the specified start time.
3. If a reminder was configured, the system will automatically dispatch the reminder emails at the specified reminder time.

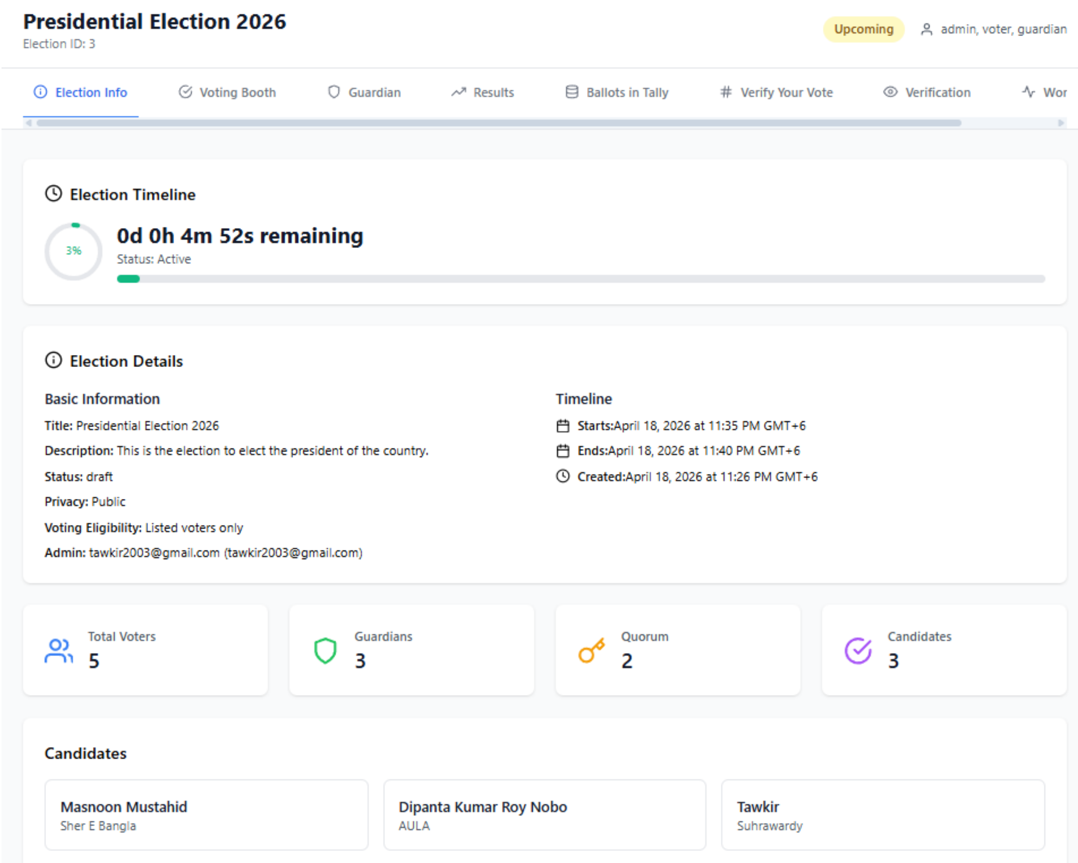


Figure 5.7: Election page confirming the scheduled start and end times after activation

Tip

After activation, share the election link with your voters so they can bookmark it. The link is visible on the election's main page and is also included automatically in the reminder email body if you enabled one.

5.4 Editing the Voter List

After an election has been created and scheduled, the **Election Admin** (the user who created the election) and any **assigned guardians** may still modify the voter list —

but only *until the election start time*. Once voting begins, the voter list is locked and can no longer be changed.

> **Note**
>
> **Who can edit the voter list?** The Election Admin and guardians assigned to this election. Both roles have the same voter-list editing privileges, and both are subject to the same deadline: all changes must be made before the scheduled election start time.

5.4.1 Accessing the Voter List Editor

1. Log in to AmarVote with an Election Admin or guardian account.
2. Navigate to **All Elections** and open the election you wish to manage.
3. On the election page, locate the **Edit Voter List** button.

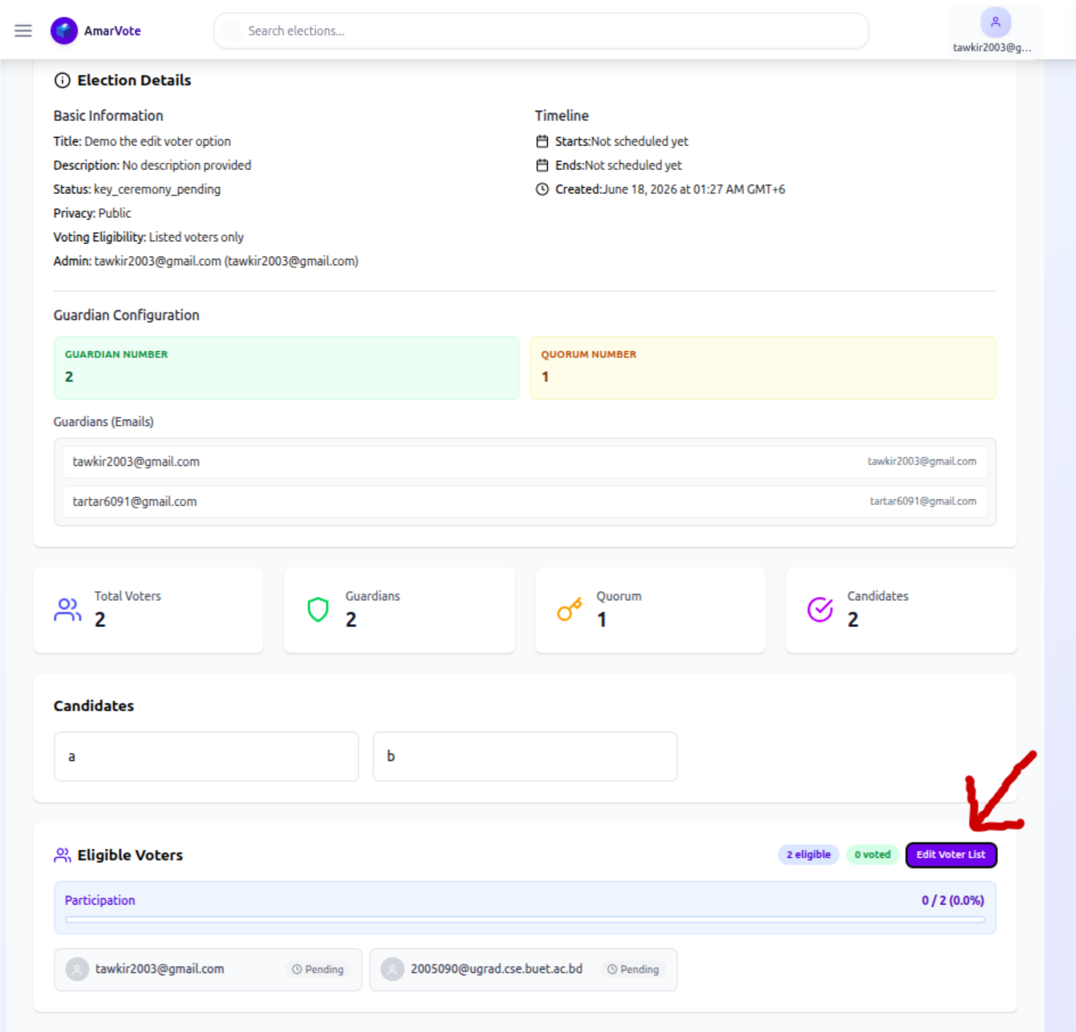


Figure 5.8: The election management page — click *Edit Voter List* to open the voter list editor

5.4.2 Adding and Removing Voters

The voter list editor allows you to add new voters or remove existing ones at any time before the election begins.

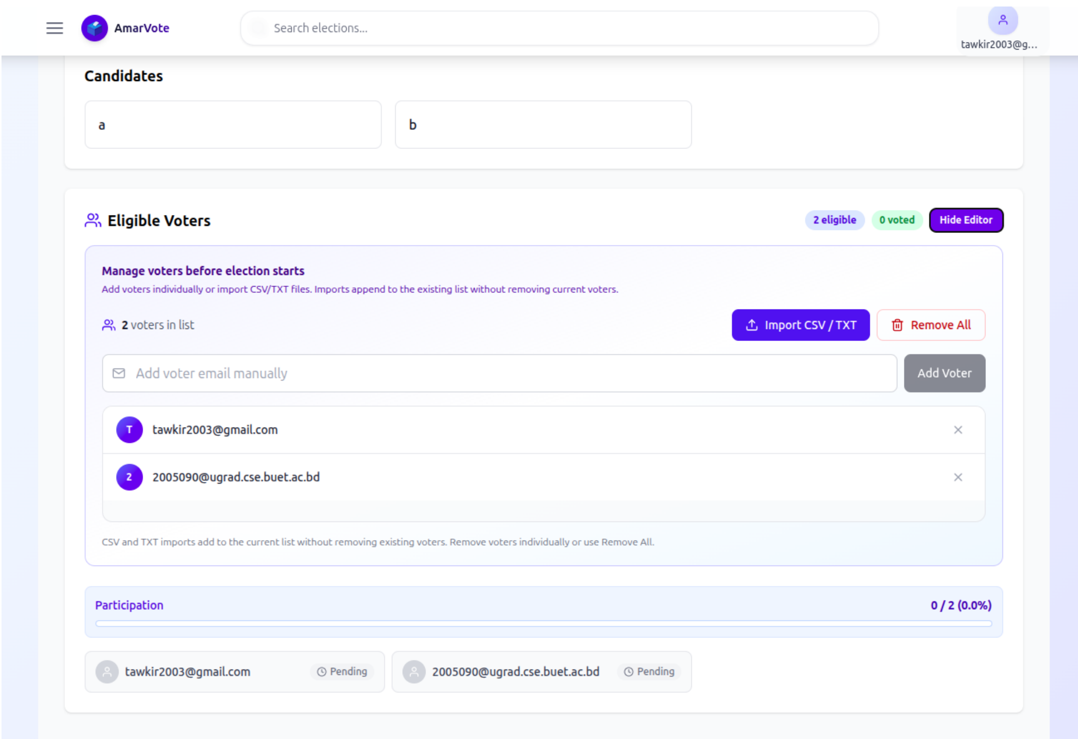


Figure 5.9: The voter list editor — add voters manually or import a CSV file; remove voters using the cross button next to each email

Removing a Voter

To remove a voter from the list:

1. In the voter list editor, find the voter's email address in the list.
2. Click the **cross (×) button** on the right side of the field containing that voter's email.
3. The voter will be removed from the list immediately.

Adding Voters Manually

To add a single voter by email:

1. In the voter list editor, locate the `Add Voter Email Manually` field.
2. Enter the voter's email address.
3. Click **Add Voter** to add them to the list.

Adding Voters via CSV Upload

To add multiple voters at once:

1. Prepare a CSV or plain-text file with one email address per row (same format as during election creation).
2. In the voter list editor, click **Import CSV/TXT**.
3. Select your prepared file from your computer.
4. The system will import all valid email addresses and add them to the voter list.

Important

Deadline: Voter list editing is only available before the election start time. Once the election opens for voting, the voter list is permanently locked — no further additions or removals are possible.

Tip

Voters added to the list who do not yet have an AmarVote account will need to register before they can vote. Consider notifying newly added voters in advance, especially if the election start time is approaching.

Part III

The Voting Process

Chapter 6

Voting — Casting Your Encrypted Ballot

AmarVote uses an **end-to-end verifiable** voting protocol. Your ballot is encrypted *before* it is stored in the database in encrypted form, the database never stores your vote in plaintext.

6.1 Accessing the Voting Booth

1. Log in to AmarVote during the election period.

2. Open the navigation menu and navigate to **All Elections**.

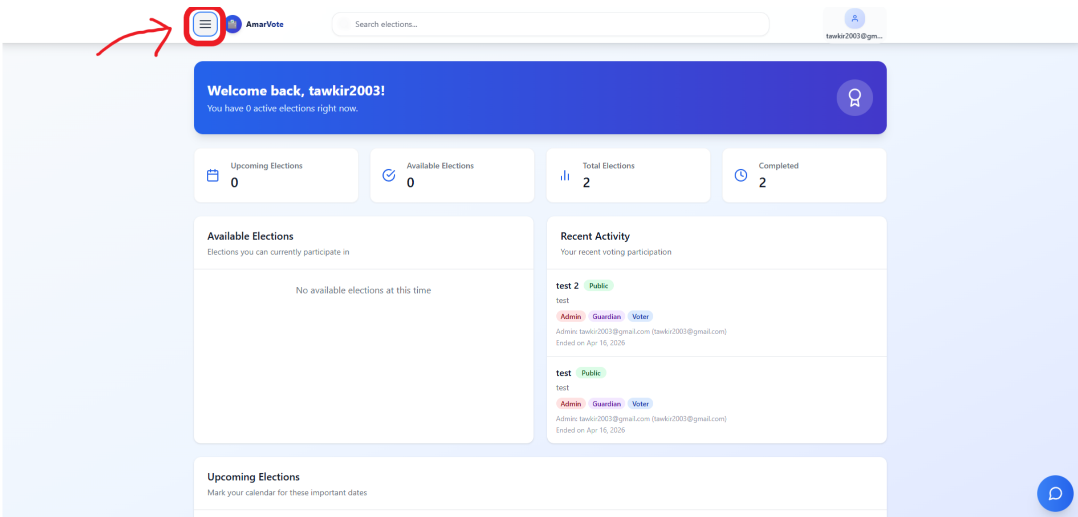


Figure 6.1: Opening the navigation menu to navigate to elections

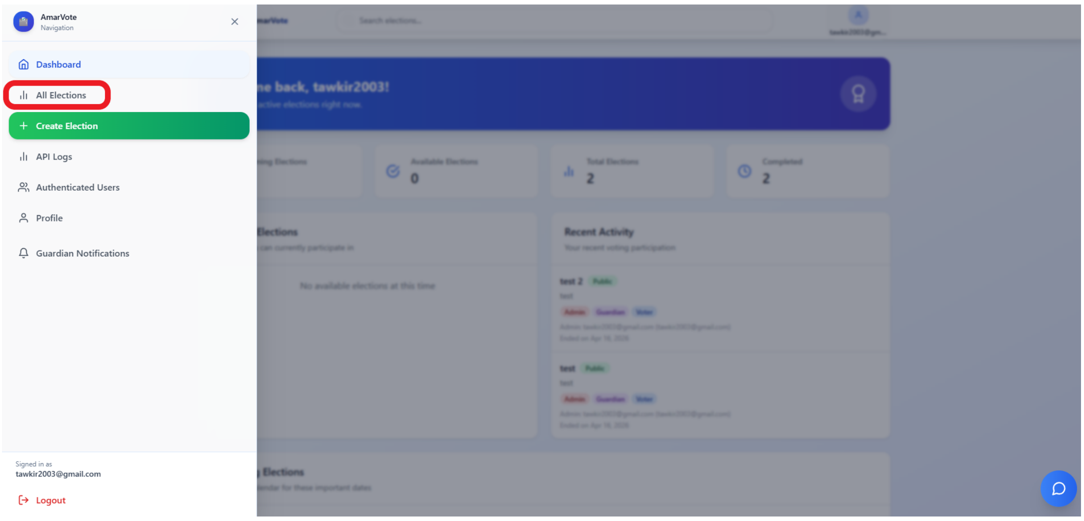


Figure 6.2: Selecting *All Elections* from the navigation menu

3. Find the election you wish to vote in. You can use the **Ongoing** filter to quickly locate active elections.

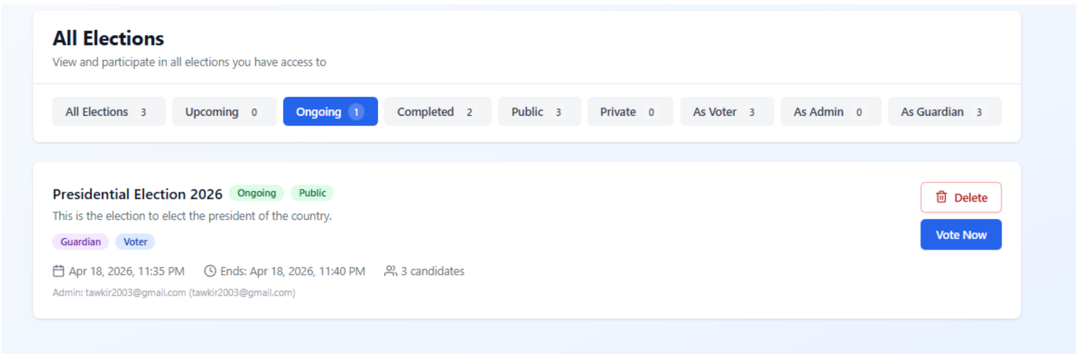


Figure 6.3: Filtering elections by *Ongoing* status to find a current election

Alternatively, active elections may also appear in your **Dashboard**.

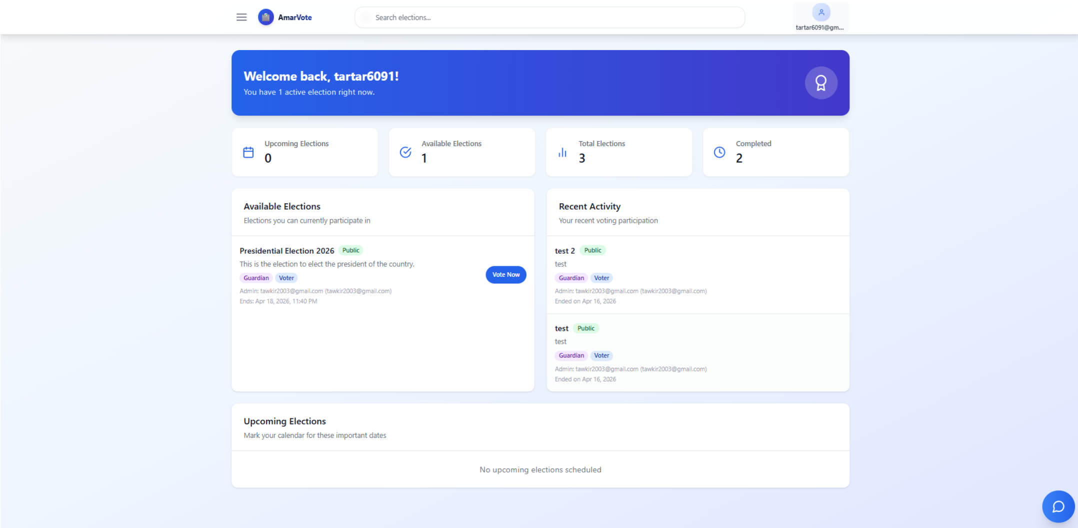


Figure 6.4: Finding current elections directly from the Dashboard

4. Open the election page by clicking on the Election title you want to vote in.
5. Click the **Voting Booth** tab.

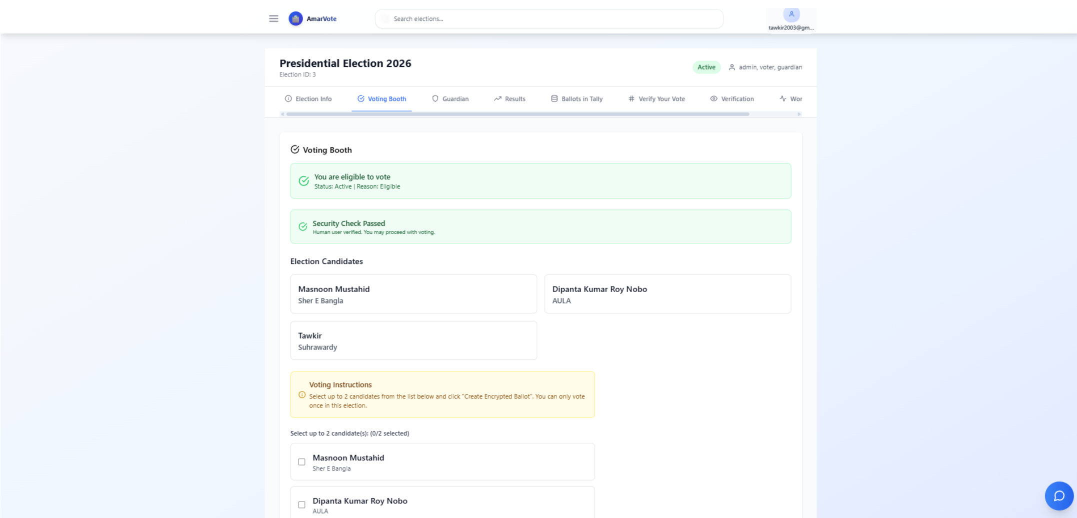


Figure 6.5: The Voting Booth tab, confirming voter eligibility before ballot creation

Important

The **Voting Booth** tab will only appear if:

- The election is currently open (between the start and end times).
- Your email is on the approved voter list (or the election is open to all users).
- You have not already cast your final ballot.

6.2 Selecting Your Candidate(s)

1. The voting booth displays all candidates with their names and photos (if uploaded).

2. Select the candidate(s) you wish to vote for. The interface will show how many you may select (e.g., "Select up to 2 candidates").
3. A checkmark will appear next to each selected candidate.
4. Once you have made your selection(s), click **Create Encrypted Ballot**.

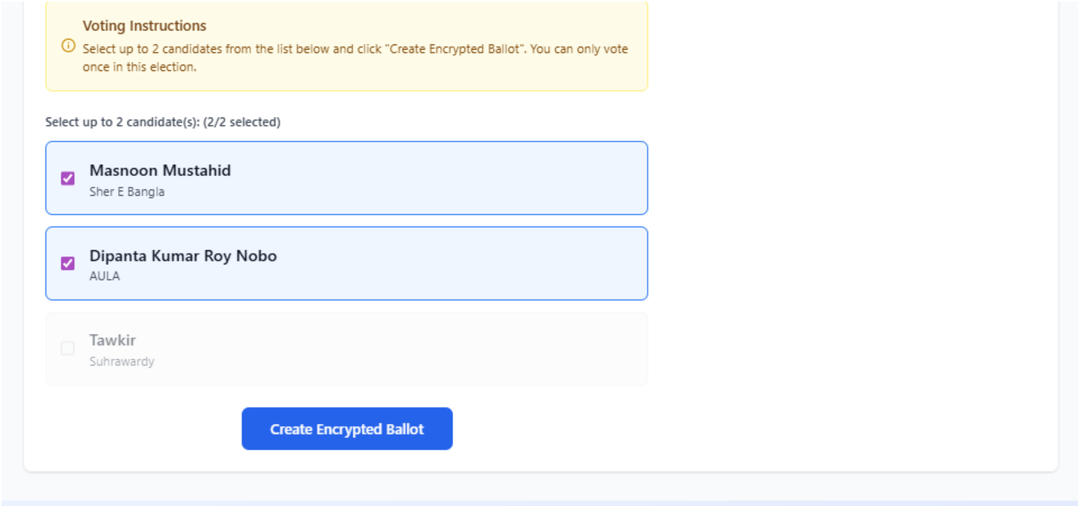


Figure 6.6: Selecting candidates and creating an encrypted ballot in the Voting Booth

> **Note**
>
> **Important distinction:** Creating an encrypted ballot is *not* the same as casting your vote. This step only creates a cryptographic envelope for your choice. You have two options from here: **cast** the ballot or **challenge** it (Benaloh Challenge).

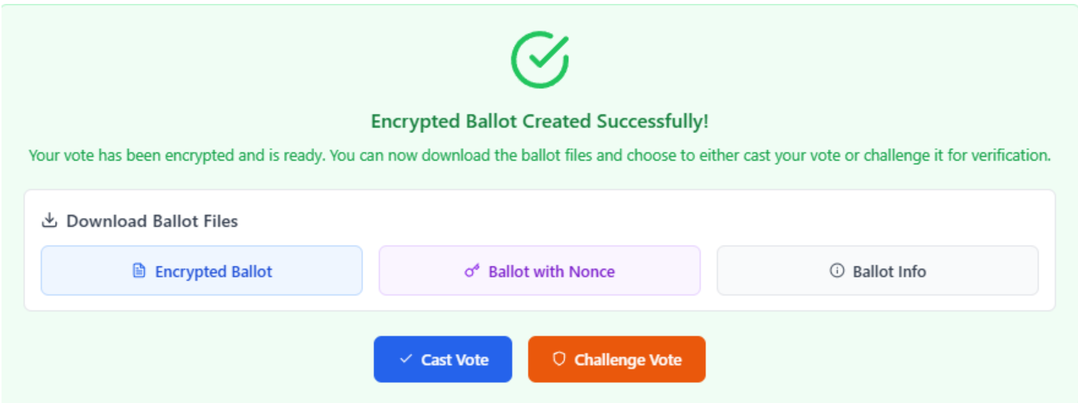


Figure 6.7: Confirmation that the encrypted ballot has been successfully created

6.3 The Benaloh Challenge

The **Benaloh Challenge** is a cryptographic audit mechanism that allows you to *verify* that the encrypted ballot accurately represents your choice — without decrypting and revealing your vote.

6.3.1 How It Works

When you create an encrypted ballot for candidates A and B:

1. The system generates an encrypted representation of your selection.
2. The challenge asks you to *re-enter* your intended vote selection.
3. The system checks whether the re-entered selection matches the encrypted ballot.
4. If it matches, you receive confirmation that the encryption is correct.
5. A mismatch is flagged — this occurs if you entered wrong candidates or if the encryption is faulty (which should never happen in a properly functioning system).

> **Note**
>
> **Key property:** If you voted for A and B, only submitting {A, B} in the challenge will return a positive result. Submitting {A}, {B}, {A, C}, etc., will return a negative result. This proves the encryption without revealing the actual vote to the server.

6.3.2 Performing the Benaloh Challenge

1. After creating an encrypted ballot, click **Challenge Vote** (Benaloh Challenge).
2. A selection interface will appear. Select what you believe you voted for.

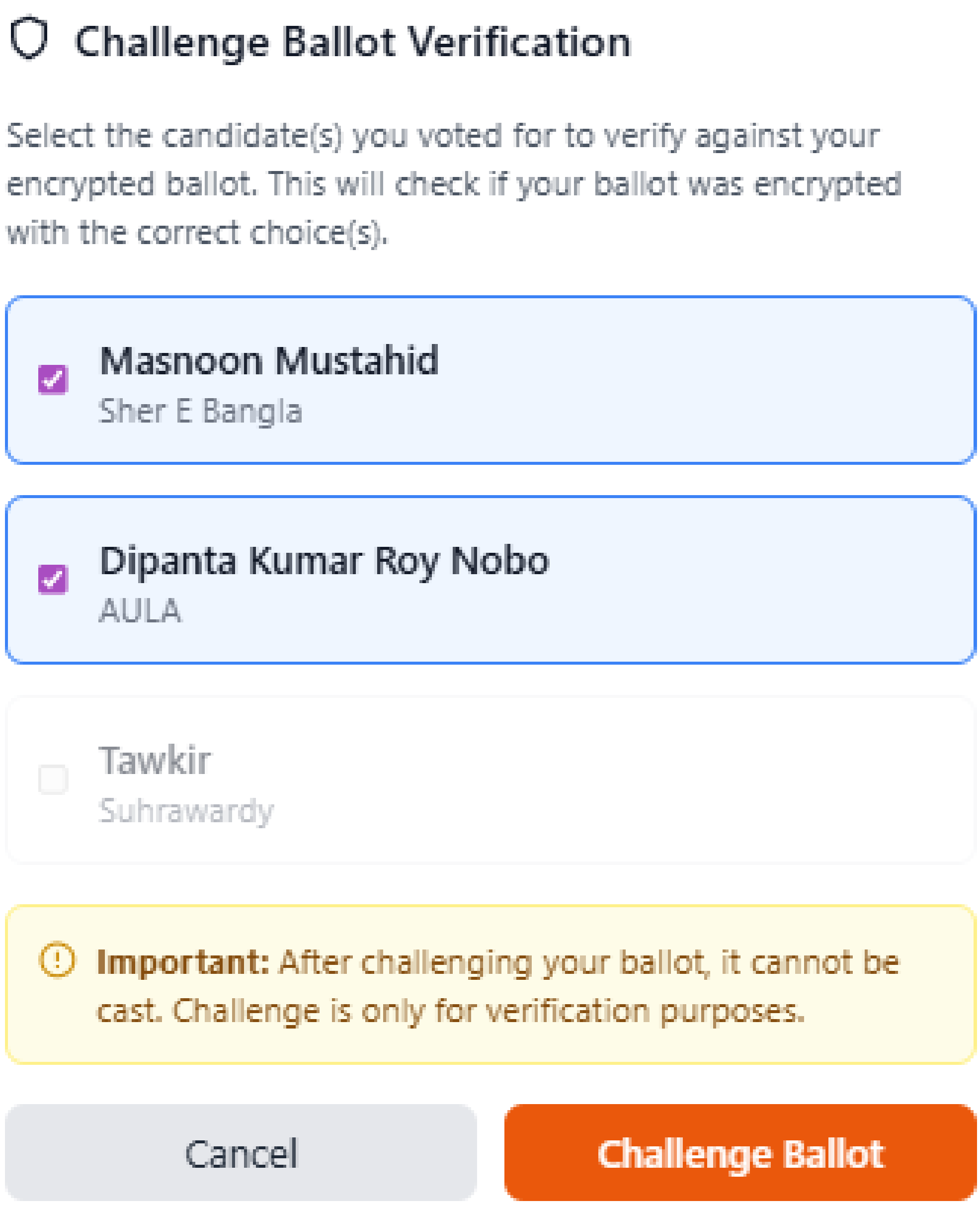
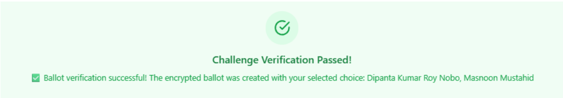
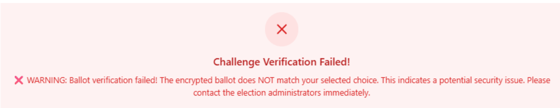


Figure 6.8: Benaloh Challenge — entering the correct candidate selection to verify the encryption

3. Click **Submit Challenge**.
4. The system will report whether the challenge passed or failed.



(a) Challenge passed — encryption verified successfully



(b) Challenge failed — encryption mismatch detected

Figure 6.9: Benaloh Challenge outcomes: passed (left) and failed (right)

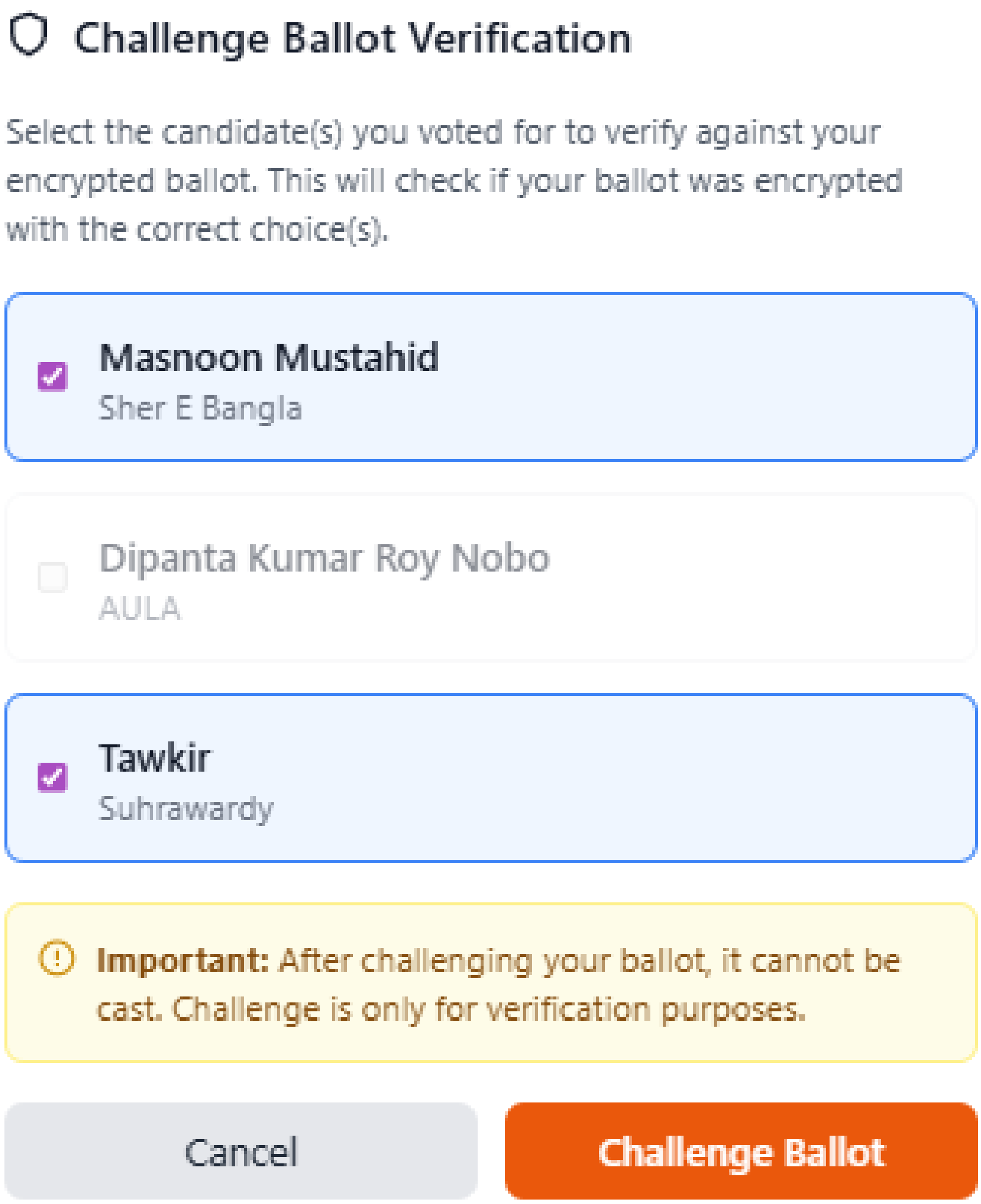


Figure 6.10: Benaloh Challenge — example of an incorrect candidate selection that causes the challenge to fail

Warning

Critical Rule: You may only perform the Benaloh Challenge **once** per encrypted ballot. Regardless of whether the challenge passes or fails, the challenged encrypted ballot is **permanently discarded**. You must then:

1. Re-select your candidates.
2. Create a new encrypted ballot.
3. Decide again: cast it or challenge it.

Tip

No limit on challenges: You may create an encrypted ballot, challenge it, discard it, and create another one as many times as you wish. This is by design — it allows you to verify the system's integrity without fear of running out of opportunities. However, you must eventually *cast* (not challenge) a ballot to have your vote counted.

6.4 Casting Your Vote

When you are satisfied that the system is working correctly and ready to cast your final vote:

1. Re-select your candidate(s) in the voting booth.
2. Click **Create Encrypted Ballot**.
3. Click **Cast Vote** (do *not* click Challenge this time).

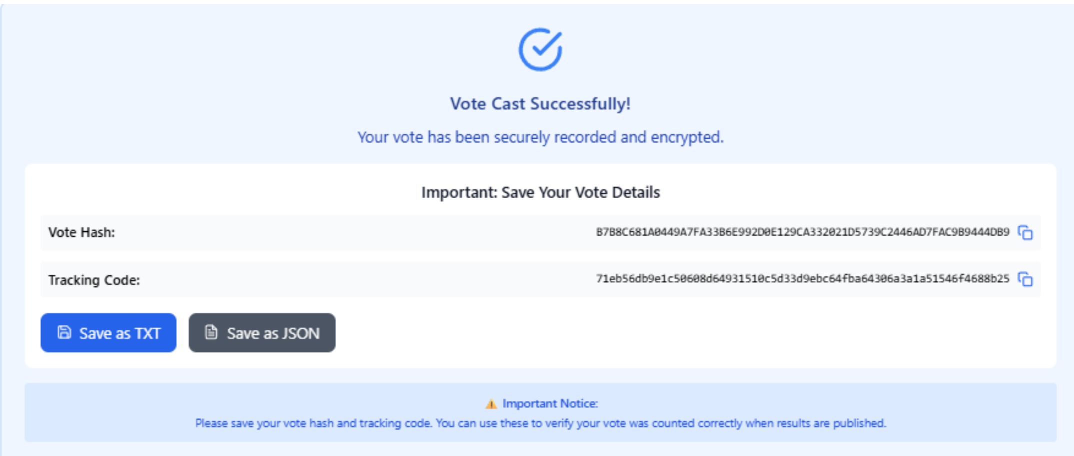


Figure 6.11: Casting the final encrypted ballot — the irreversible submission step

Warning

This action is irreversible. Once you cast your ballot, it cannot be changed, recalled, or re-cast. Your vote is final.

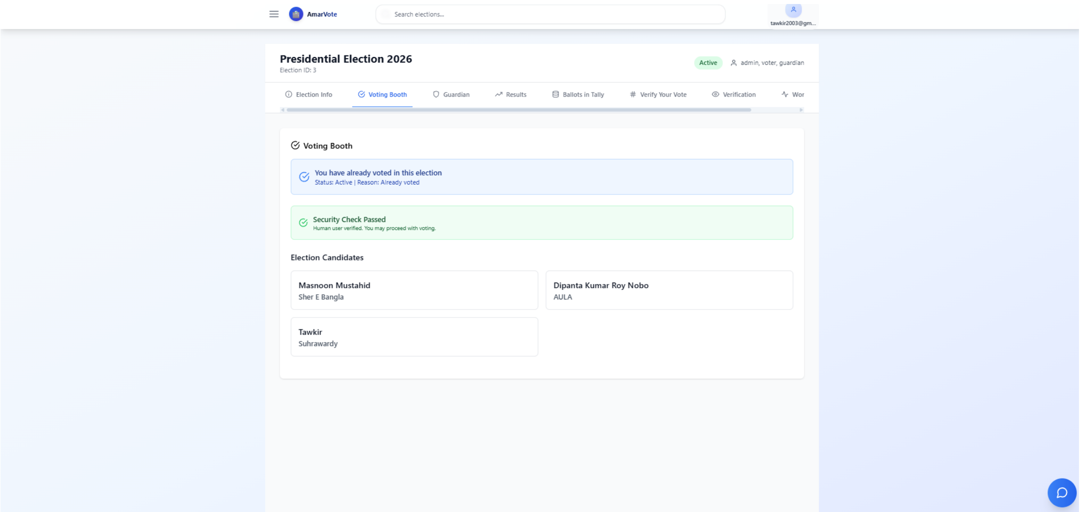


Figure 6.12: System confirmation that a voter cannot cast more than one final ballot

6.5 Your Voter Receipt

After successfully casting your ballot, AmarVote will send you an email containing your **Voter Receipt**:

- **Ballot Tracking Code**: A unique identifier for your ballot in the tally system.
- **Ballot Hash**: A cryptographic fingerprint of your encrypted ballot.

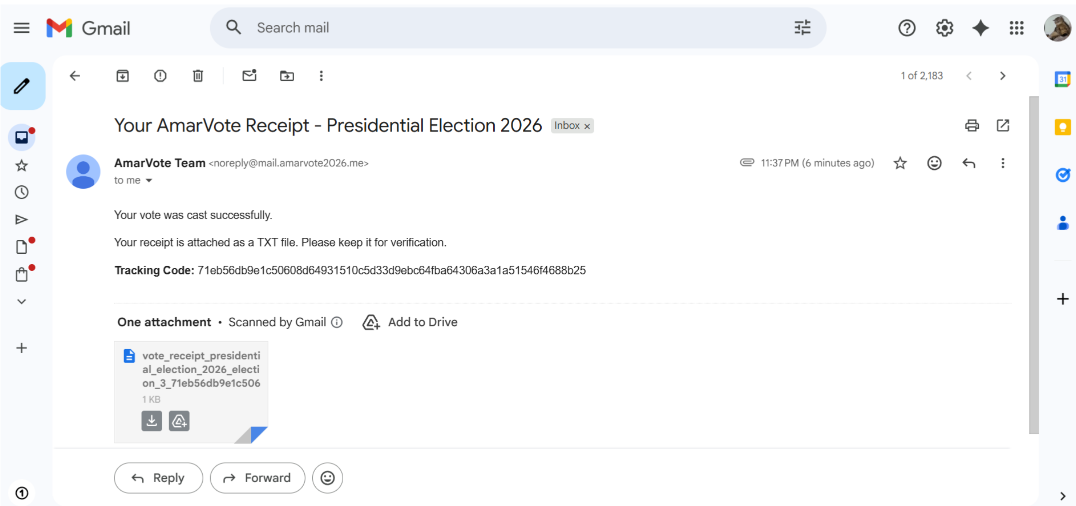


Figure 6.13: Voter receipt email received after successfully casting an encrypted ballot

Tip

Save your receipt! You will need both the tracking code and the hash to verify your vote after the tally is complete. Save the email or note these values in a secure place.

Part IV

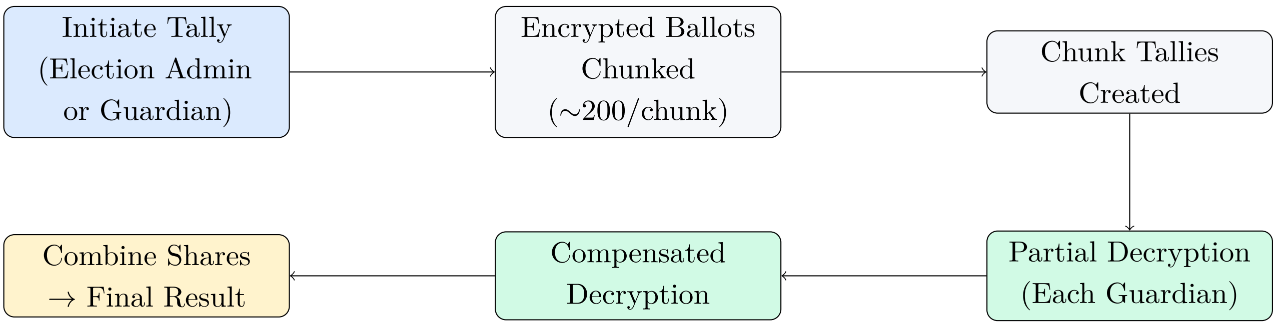
Tallying & Decryption

Chapter 7

The Tallying Process

After the election closes, the encrypted ballots must be tallied and then decrypted by the guardians to reveal the final results.

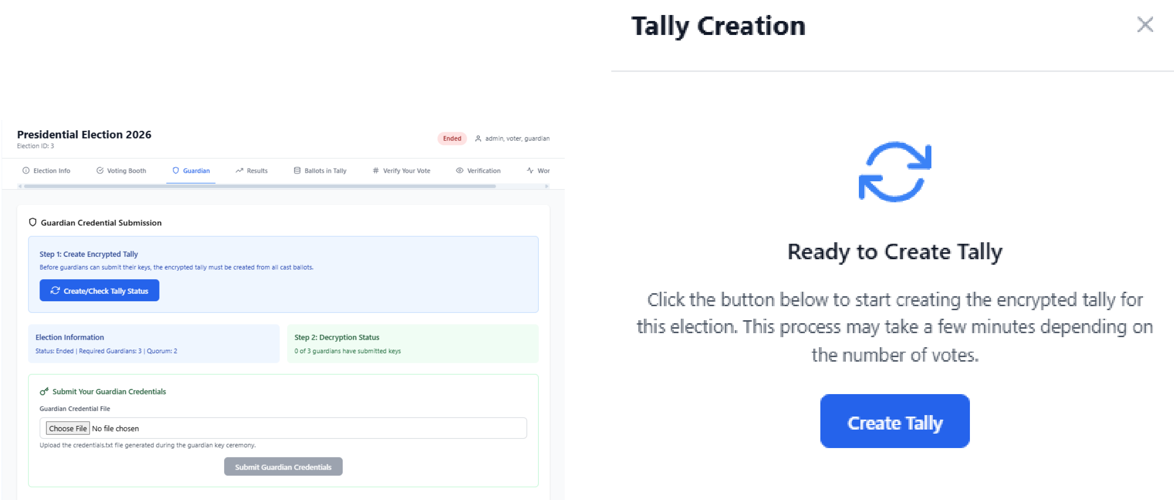
7.1 Overview of the Tally Pipeline



7.2 Initiating the Tally

Either the Election Admin or any guardian may start the tally process.

1. Navigate to the election page after the election has closed.
2. In the upper tab, click **Create/Check Tally Status**.
3. If the tally has not yet been created, you will see a **Create Tally** button. Click it to start the tallying process.
4. The system will begin processing. This involves:
   - Retrieving all cast encrypted ballots from the database.
   - Dividing them into chunks of approximately 200 ballots each.
   - Computing an encrypted tally for each chunk using homomorphic addition.
   - Store each of the tally chunks back to the database for later decryption.
5. A progress indicator will display the number of chunks processed vs. total.
6. When complete, a **"Tally Complete"** confirmation message is displayed.



(a) Tally creation in progress — first stage (b) Tally creation in progress — second stage

Figure 7.1: Tally creation progress stages

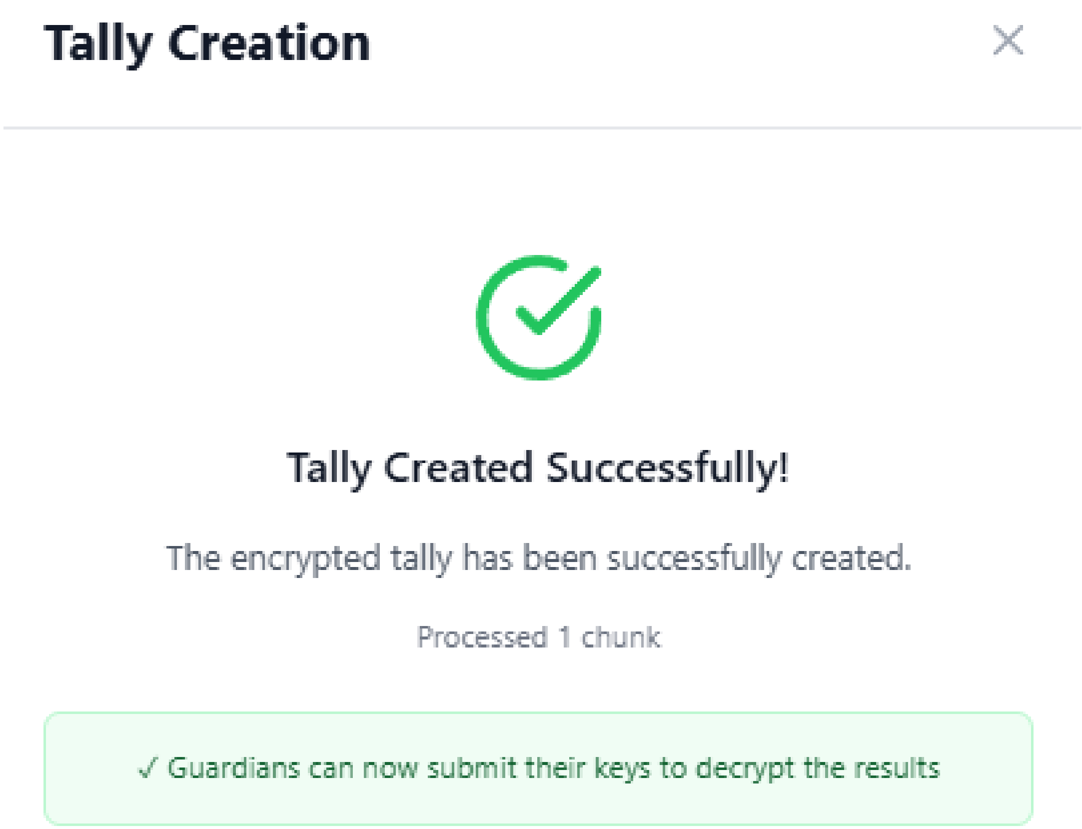


Figure 7.2: Confirmation that the encrypted tally has been created successfully

Note

The tally computation is performed asynchronously in the background. For large elections (thousands of votes), this may take several minutes. You do not need to remain on the page — progress is saved and you can return to check.

Chapter 8

Cryptographic Decryption

This is the final cryptographic phase. The encrypted tally can only be decrypted by combining a minimum of *quorum* guardian key shares.

8.1 Partial Decryption (Guardian Action)

1. Guardians are notified that the tally phase is complete and partial decryption can begin.

2. Each guardian navigates to the election page and opens the **Guardian** tab.

3. Click **Choose File** and select your private key file from Phase 1.

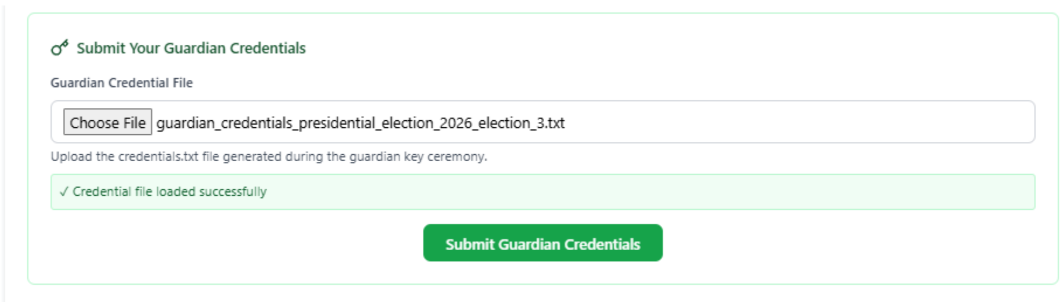


Figure 8.1: Uploading the encrypted private key file to begin partial decryption

4. Click **Submit Guardian Credentials**.

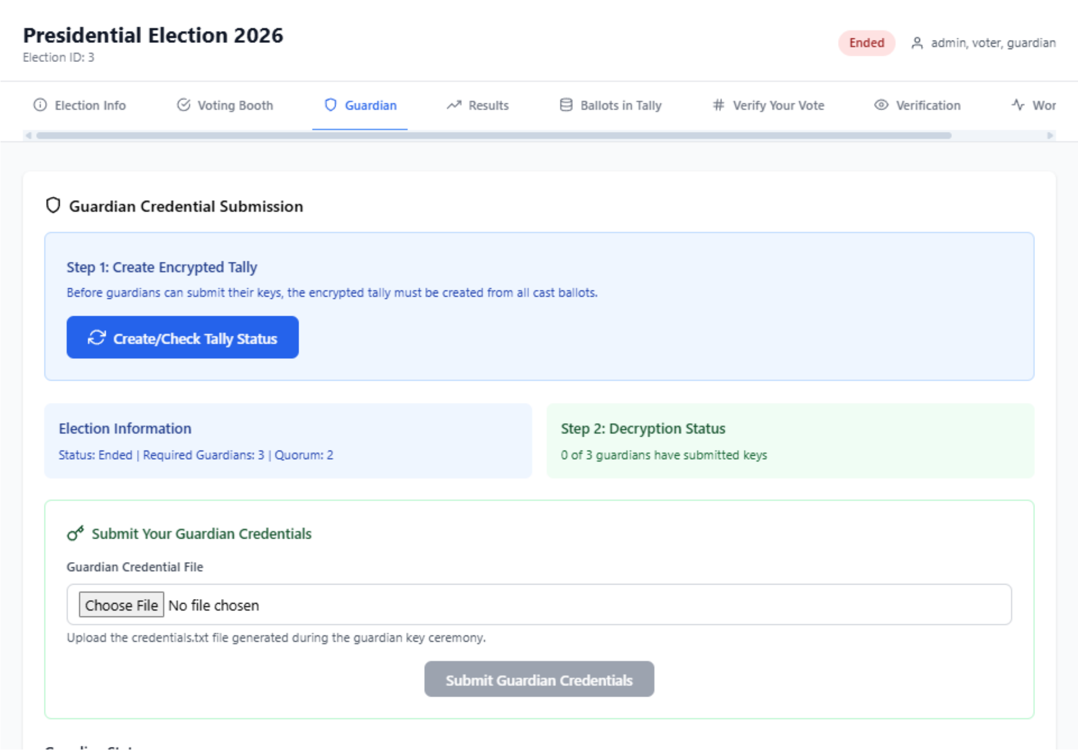


Figure 8.2: Submitting guardian credentials to initiate partial decryption

5. The system will:

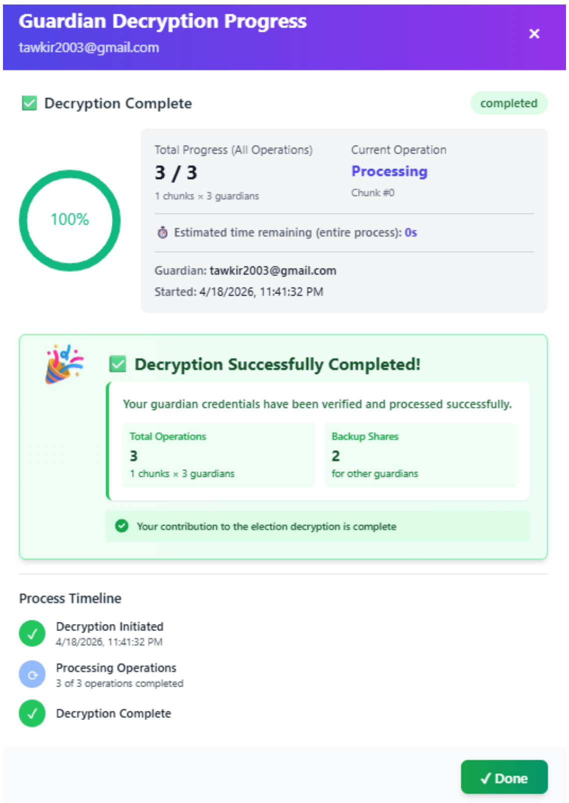
(a) Decrypt your private key using your password (stored in database encrypted by ML KEM 1024).

(b) Apply your private key to each tally chunk — producing **partial decryptions** for your share.

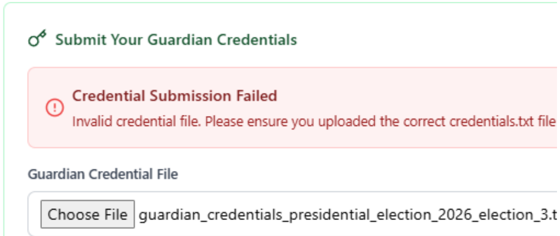
(c) Generate **compensated decryptions** for all other guardians (using the encrypted backup shares from Phase 2, decrypted with your private key). This allows the system to substitute for absent guardians.

(d) Store all partial and compensated decryptions to the database.

6. A confirmation message is displayed when your decryption contribution is submitted.



(a) Successful partial decryption submission



(b) Error: incorrect or mismatched credential file

Figure 8.3: Guardian decryption outcomes: success (left) and credential error (right)

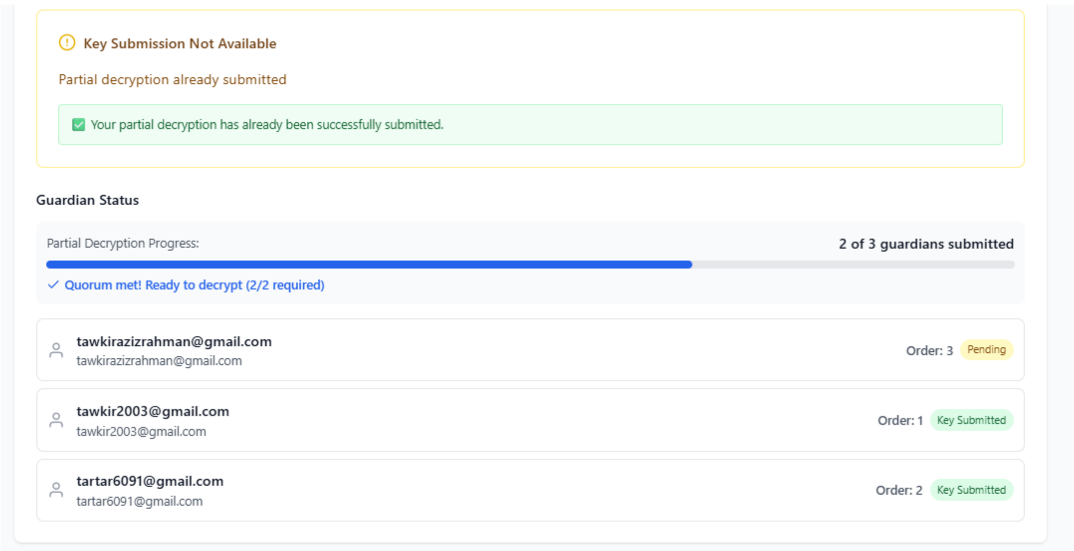


Figure 8.4: Dashboard showing quorum progress — number of guardians who have submitted partial decryptions

> **Important**
>
> The partial decryption process may take several minutes depending on the number of tally chunks. You can close the browser tab while decryption is running.

8.2 Combining Decryption Shares (Election Admin or Guardian)

Once at least *quorum* guardians have submitted their partial decryptions:

1. Navigate to the election's **Results** tab.
2. The **Combine Partial Decryptions** button will become active.
3. Click **Combine Partial Decryptions**.

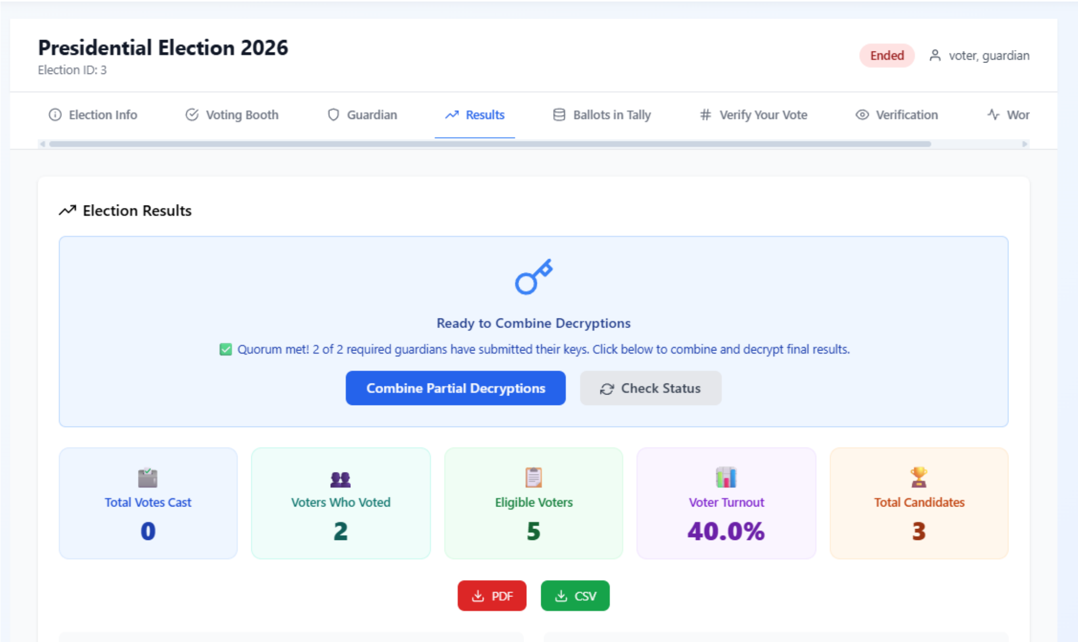


Figure 8.5: Combining decryption shares to produce the final plaintext tally

4. The system mathematically combines all submitted partial and compensated decryption shares to produce the **final plaintext tally**.
5. The election results are now computed and published.

Tip

Results are displayed as a bar chart with vote counts per candidate, alongside percentage breakdowns and total ballot counts. All cryptographic proofs are also made available for public audit.

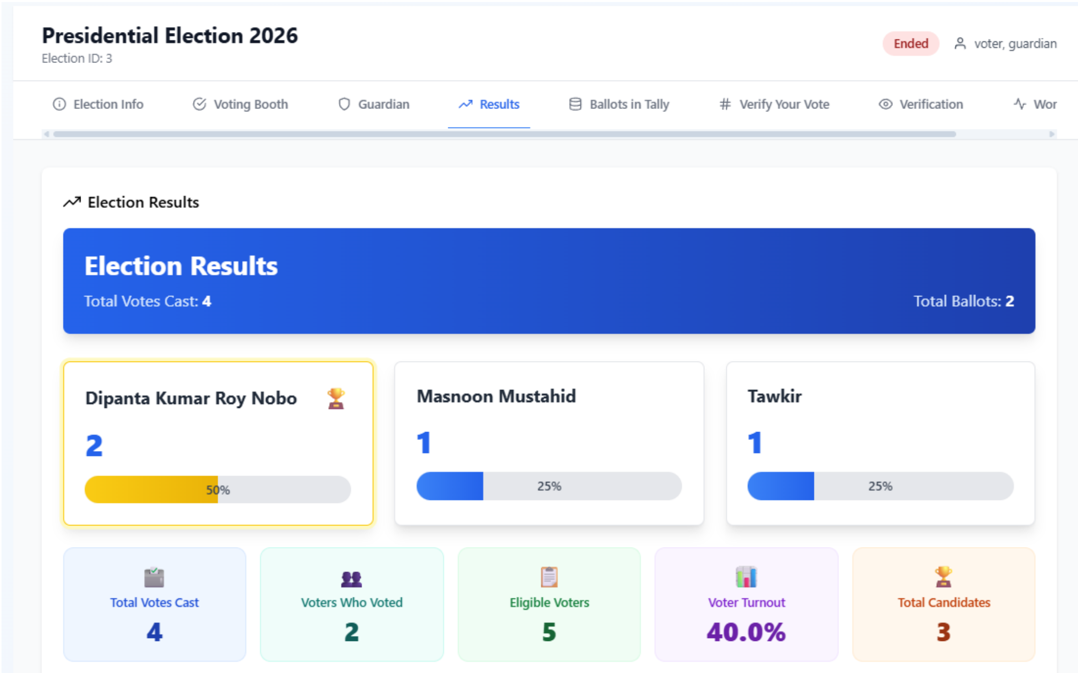
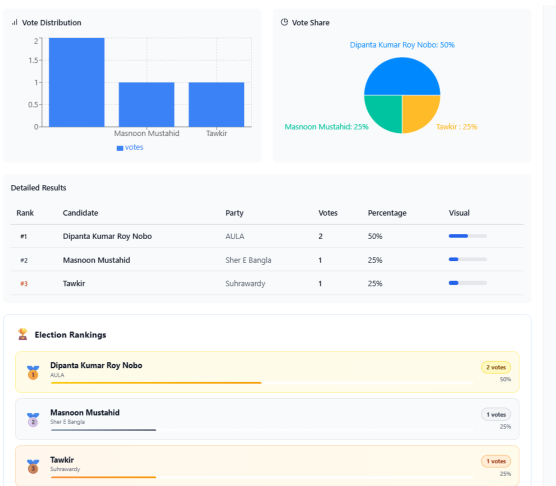
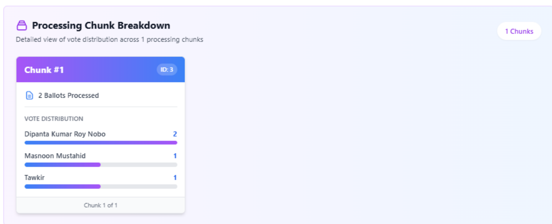


Figure 8.6: Final election results page showing vote counts and candidate standings



(a) Election results displayed as graphs and charts



(b) Detailed chunk-level tally breakdown

Figure 8.7: Election results analytics: visual charts (left) and chunk tally breakdown (right)

Part V

Verification & Audit

Chapter 9

Verifying Your Vote

AmarVote is designed to be **fully auditable**. Every voter, every observer, and every independent researcher can verify the integrity of the election.

9.1 Using Your Voter Receipt

After the results are published:

1. Navigate to the election page and open the **Verify your vote** tab.
2. Click **Choose File** and upload your ballot receipt you received in the email after your encrypted ballot was cast.

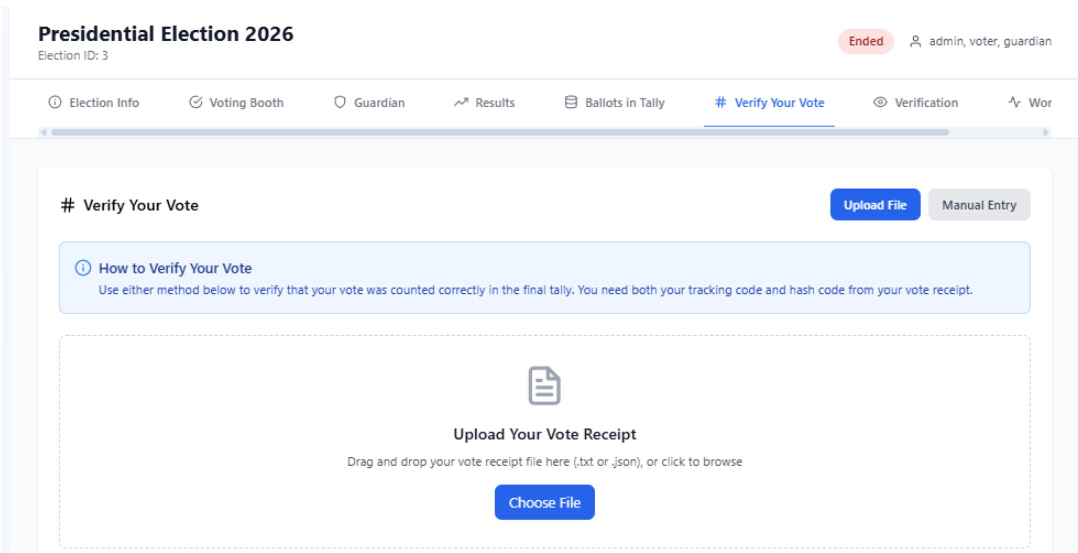


Figure 9.1: Uploading the ballot receipt file to initiate vote verification

The system will confirm:

- **Ballot Found**: Your tracking code exists in the tally, confirming your ballot was *recorded*.
- **Hash Match**: The ballot hash in the tally matches your receipt, confirming your ballot was *not modified* after submission.

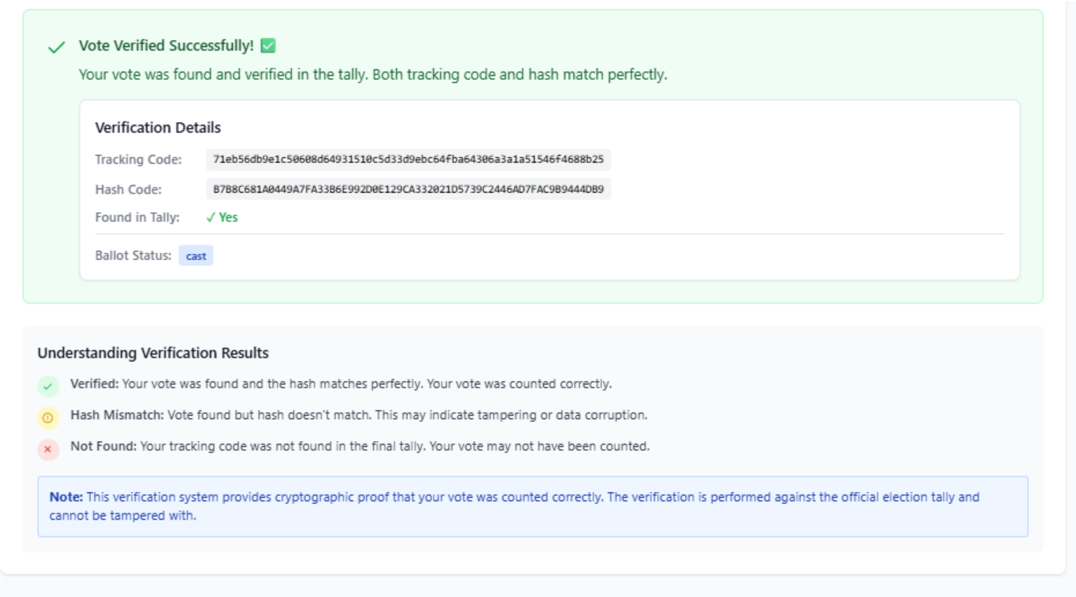


Figure 9.2: Successful ballot verification — the system confirms the ballot was recorded and unmodified

Tip

If the ballot hash does not match, this indicates potential tampering. Report this immediately to the Election Admin and your national or institutional election authority.

9.2 Manual Ballot Lookup

If you prefer to verify without relying on the automated tool:

1. Navigate to the election's **Ballots in Tally** tab.
2. This displays the complete list of all encrypted ballots included in the tally.

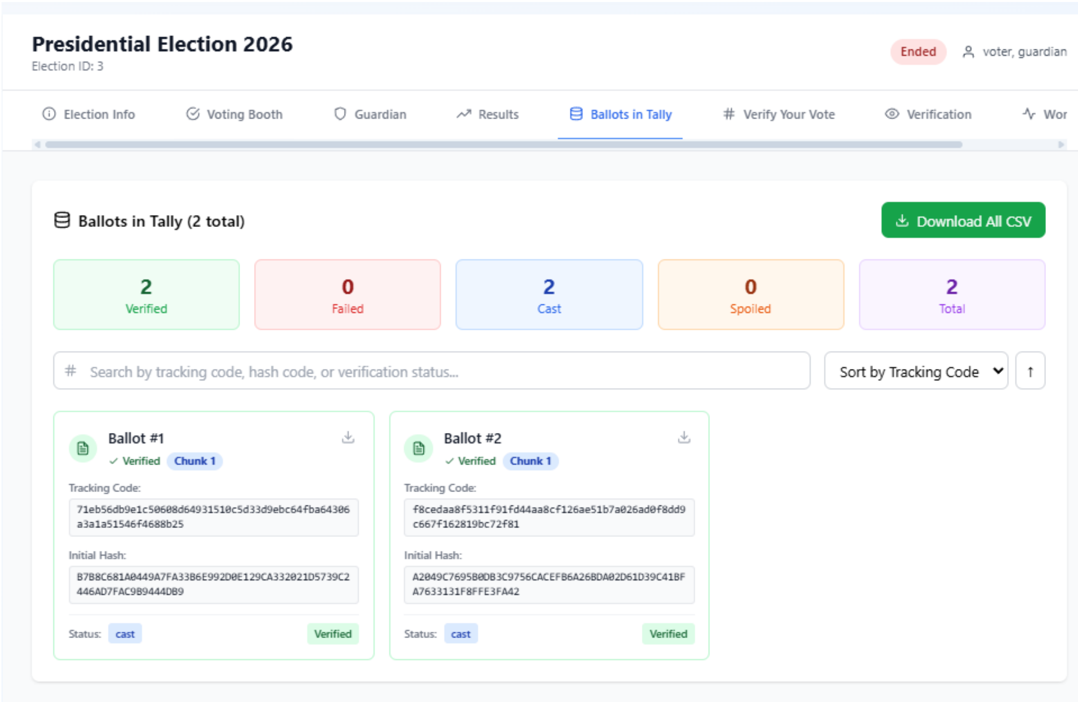


Figure 9.3: The Ballots in Tally tab, listing all encrypted ballots included in the final tally

3. Manually search through the list for your Ballot Tracking Code.
4. Compare the displayed ballot hash with the hash in your receipt email.

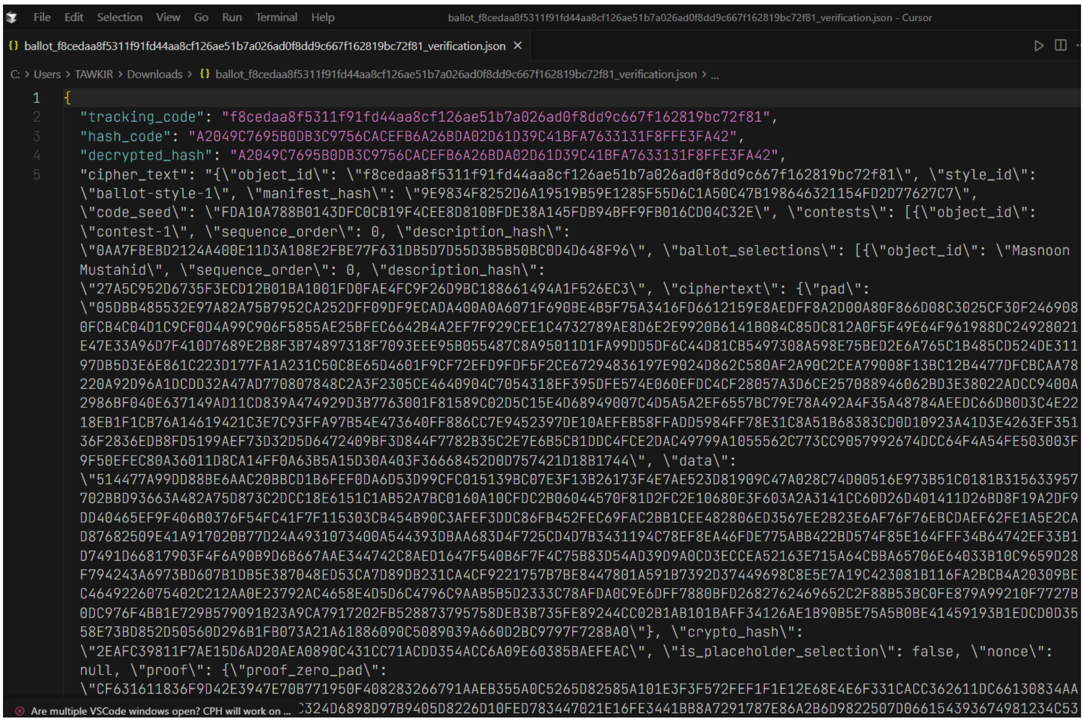


Figure 9.4: Viewing the detailed content of an individual encrypted ballot in the tally

> **Note**
>
> This manual approach is deliberately available for maximum transparency. Ballot tracking codes are anonymised — they identify the ballot but not the voter — so this list is safe to make public.

9.3 Public Election Data Download

All election data (except private keys) is publicly available for independent audit:

1. Navigate to the election's **Verification** tab.
2. You can download:
   - All encrypted ballots for chunks
   - Public keys of all guardians
   - The encrypted tally of different chunks
   - All partial decryptions and proofs of different chunks

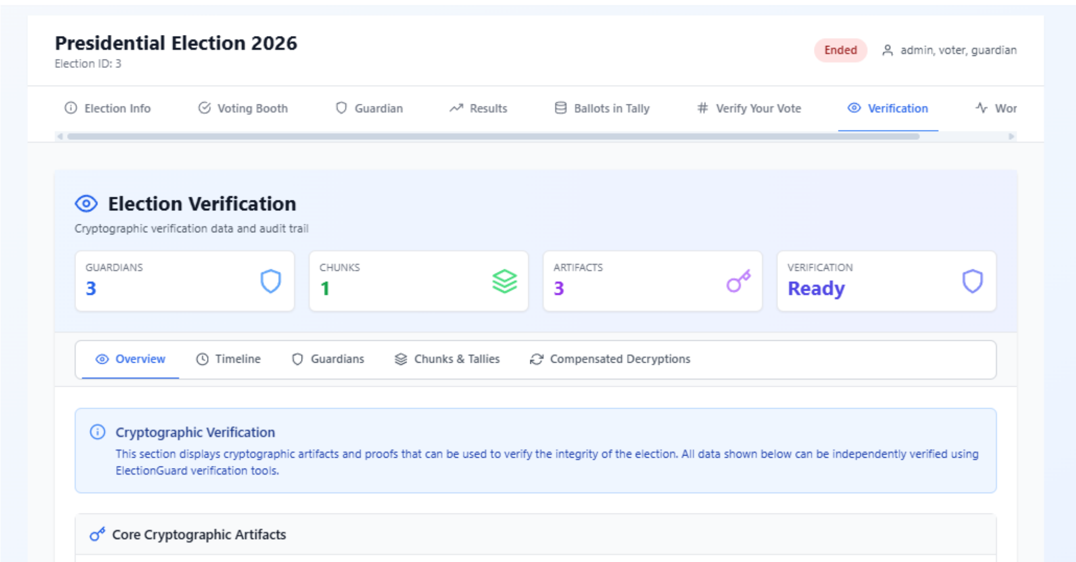


Figure 9.5: The Verification tab on the election page, available after results are published

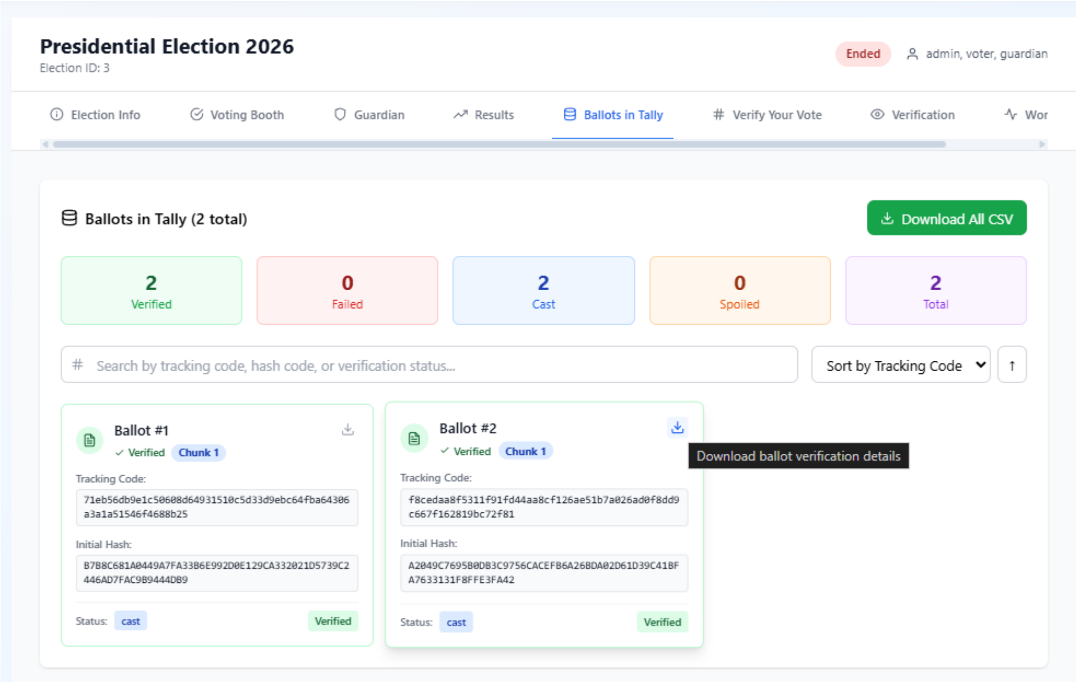


Figure 9.6: Downloading individual encrypted ballots for independent audit verification

Note

Any technically capable third party can use this downloaded data to re-verify the entire election independently using the ElectionGuard specification.

Appendix A

Frequently Asked Questions

Can the administrator see how I voted?

No. Your ballot is encrypted using the combined public keys of all guardians before it is stored. Neither the Election Admin nor any App Admin can decrypt individual ballots — only the aggregate tally can be decrypted, and only with guardian cooperation.

What if a guardian is unavailable during decryption?

If the number of available guardians meets or exceeds the quorum threshold, the election results can be published. The backup shares generated during Phase 2 are used to compensate for absent guardians.

Can I vote more than once?

No. The system enforces a one-final-ballot-per-voter policy. While you can create and challenge multiple encrypted ballots (for verification purposes), casting your vote is a one-time action.

Is my identity linked to my ballot?

Your voter receipt email contains a Ballot Tracking Code and Hash. Ballot hash is mathematically derived from the *encrypted* ballot — they do not reveal your identity or your vote. The public tally list contains tracking codes but no voter names. But for your privacy do not share the tracking code or ballot hash with anyone else, as they could be used to identify your encrypted ballot in the public tally. Though the ballot will still be encrypted, it is best practice to keep your receipt information private.

What is the Benaloh Challenge for?

It is a cryptographic proof that the encryption software is working correctly — that the encrypted ballot represents what you actually selected. It provides mathematical confidence without revealing your vote to the server.

What if I lose my guardian credentials?

Immediately notify the Election Admin. If the remaining guardians still meet the quorum threshold, tallying can proceed without you (using compensated decryption). If your loss would drop available guardians below quorum, the election results may be unrecoverable.

Appendix B

Security Architecture Summary

Component	Security Property
Ballot encryption	ElectionGuard homomorphic encryption (El Gamal based)
Private key storage	AES-256 encryption with guardian-chosen password
Key distribution	Shamir's Secret Sharing (threshold scheme)
Ballot verification	Zero-knowledge proofs included with each ballot
Guardian backup	Encrypted with each guardian's public key
Transit security	HTTPS/TLS (Nginx production configuration + Cloudflare Proxy Configuration)
Authentication	JWT tokens
Password hashing	bcrypt (Spring Security default)
Infrastructure	Docker containers, PostgreSQL, Redis, RabbitMQ